

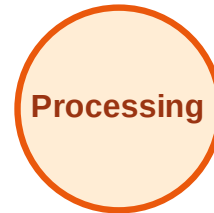
BFS Traversal

Step 1: Start at A

Legend



Complete



Processing



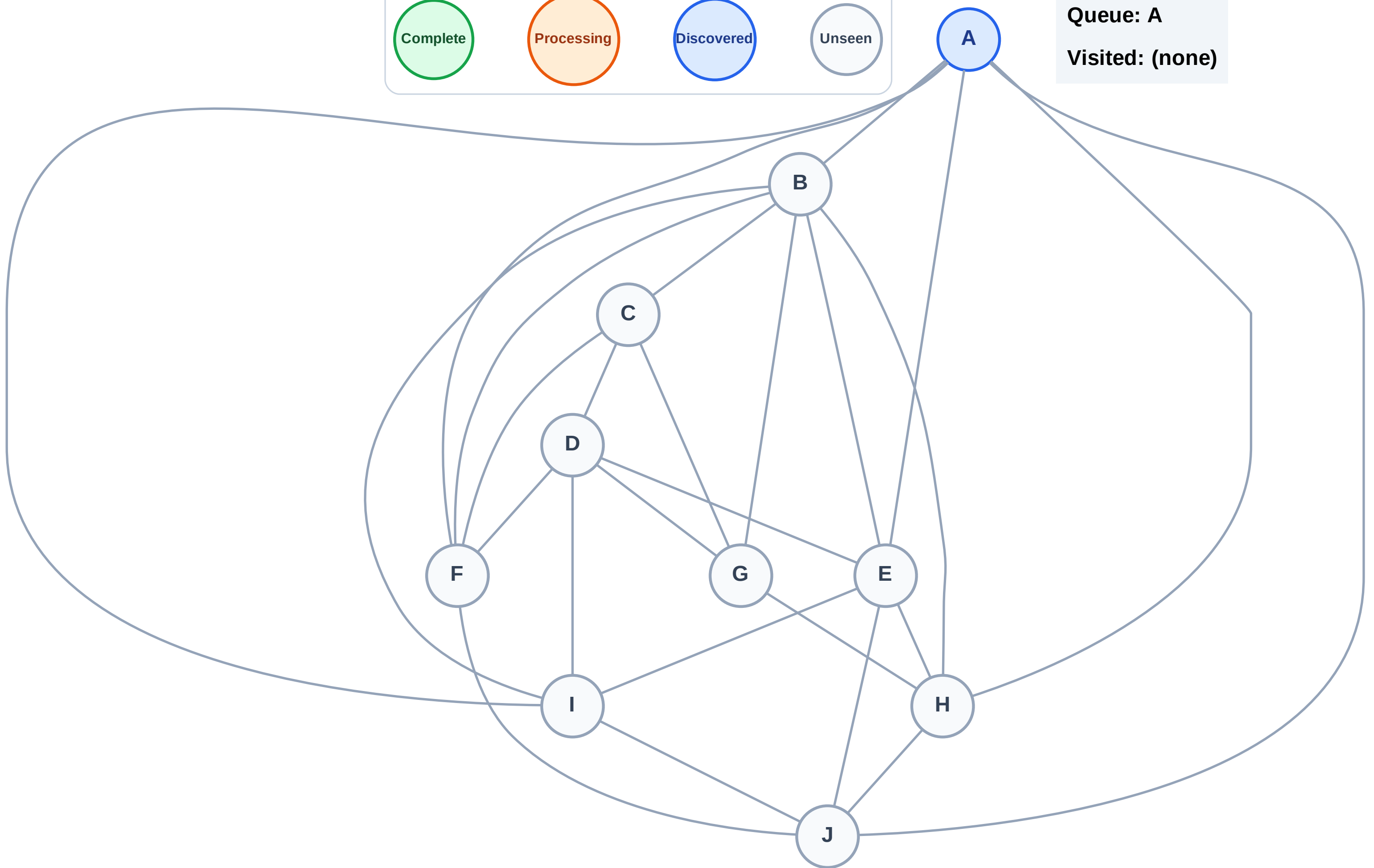
Discovered



Unseen

Queue: A

Visited: (none)



BFS Traversal

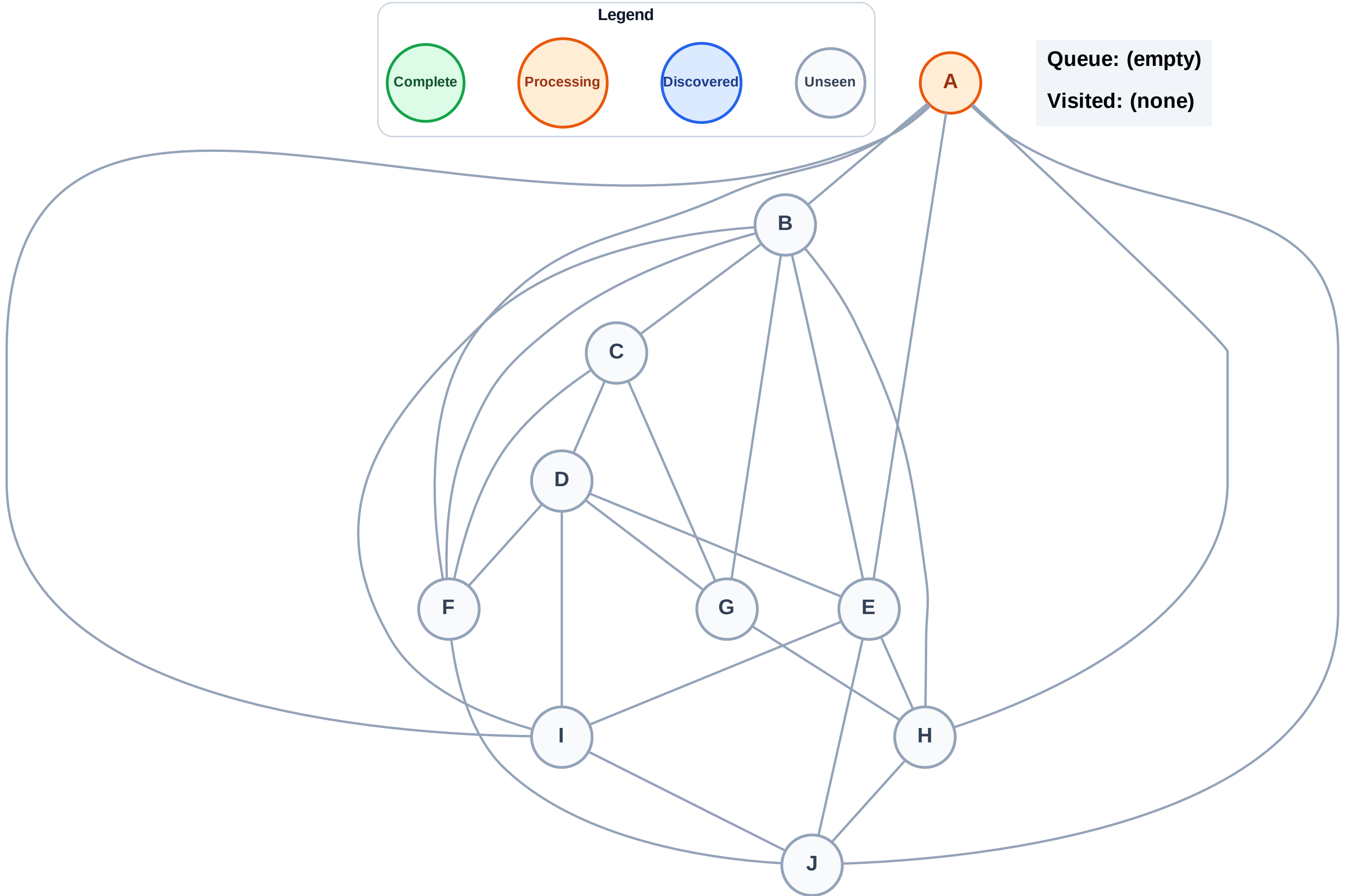
Step 2: Process node A

Legend



Queue: (empty)

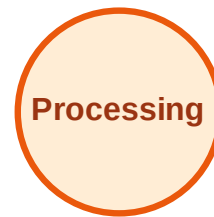
Visited: (none)



BFS Traversal

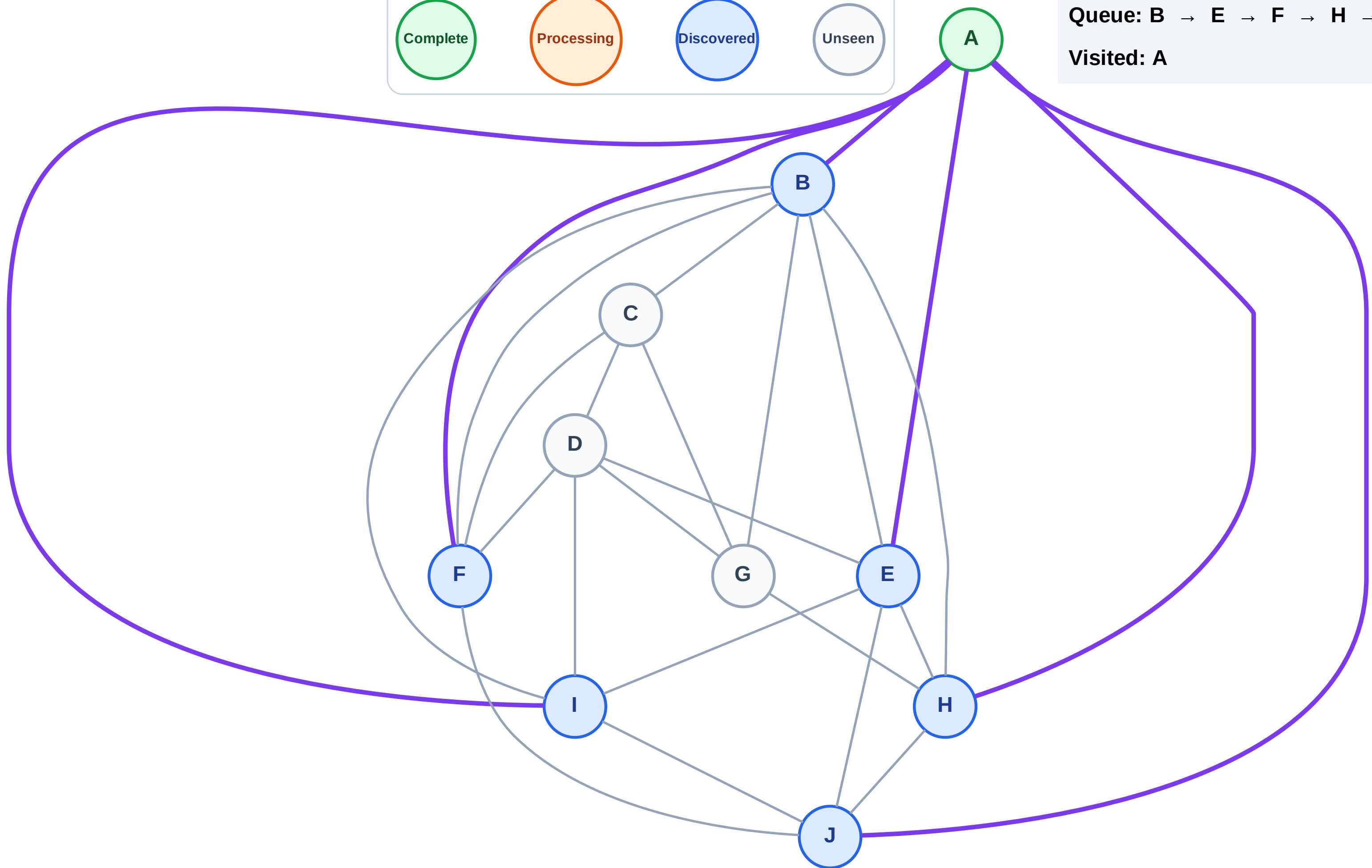
Step 3: Discover B, E, F, H, I, J from A

Legend



Queue: B → E → F → H → I → J

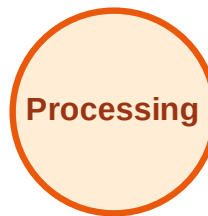
Visited: A



BFS Traversal

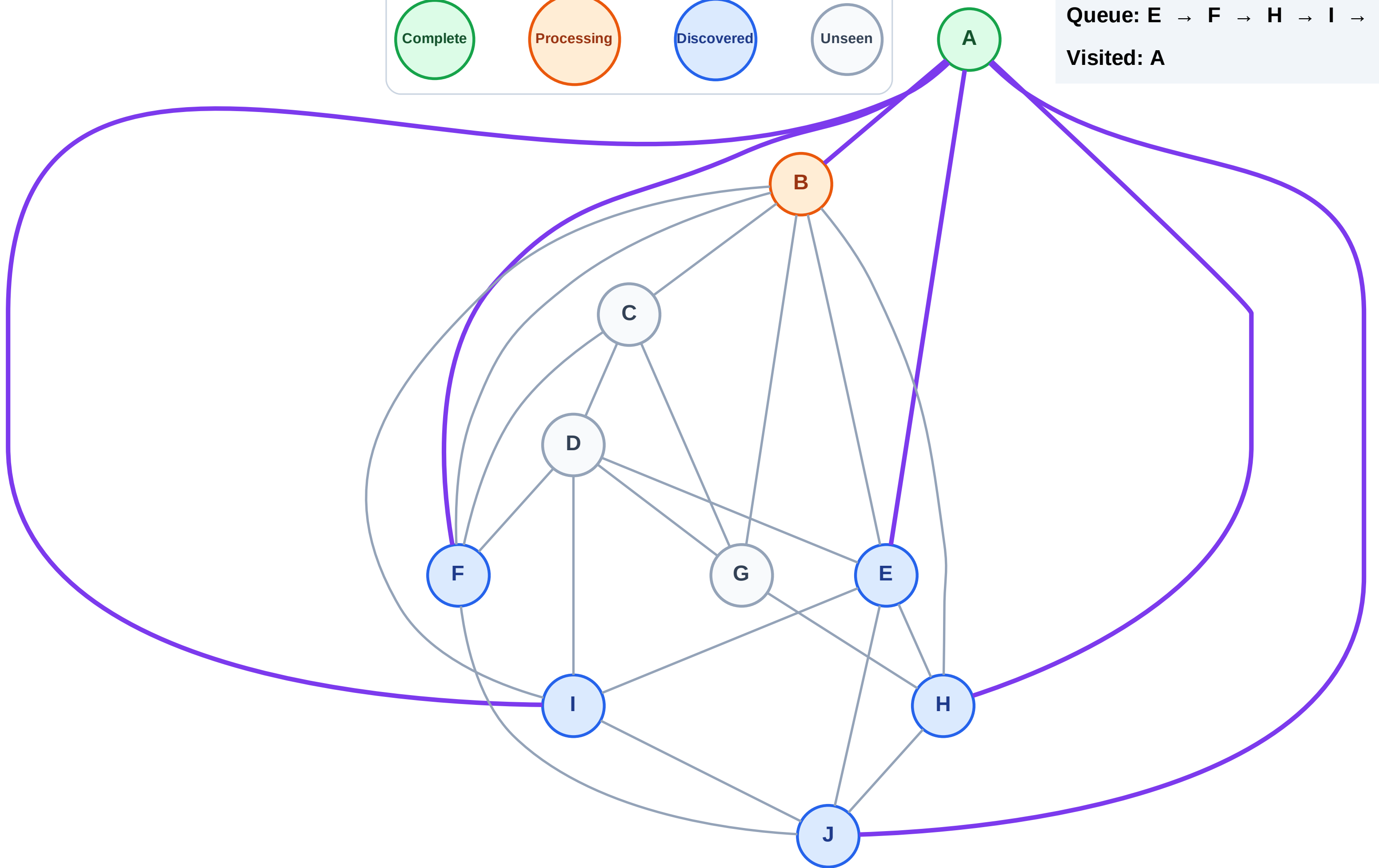
Step 4: Process node B

Legend



Queue: E → F → H → I → J

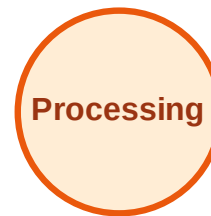
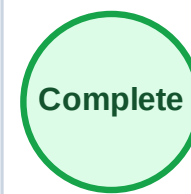
Visited: A



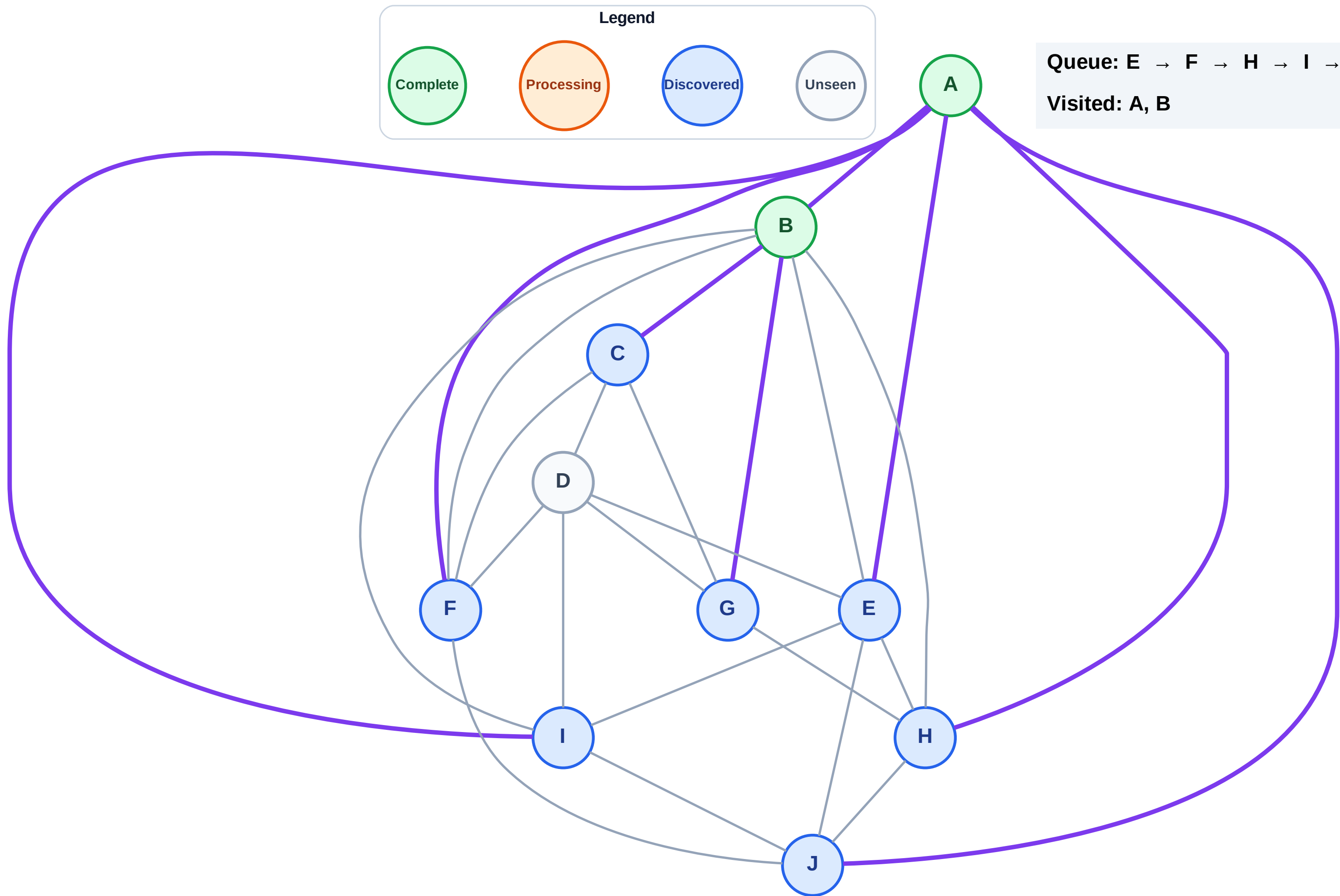
BFS Traversal

Step 5: Discover C, G from B

Legend



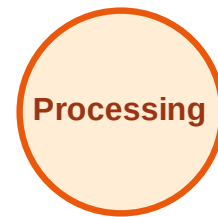
Queue: E → F → H → I → J → C → G
Visited: A, B



BFS Traversal

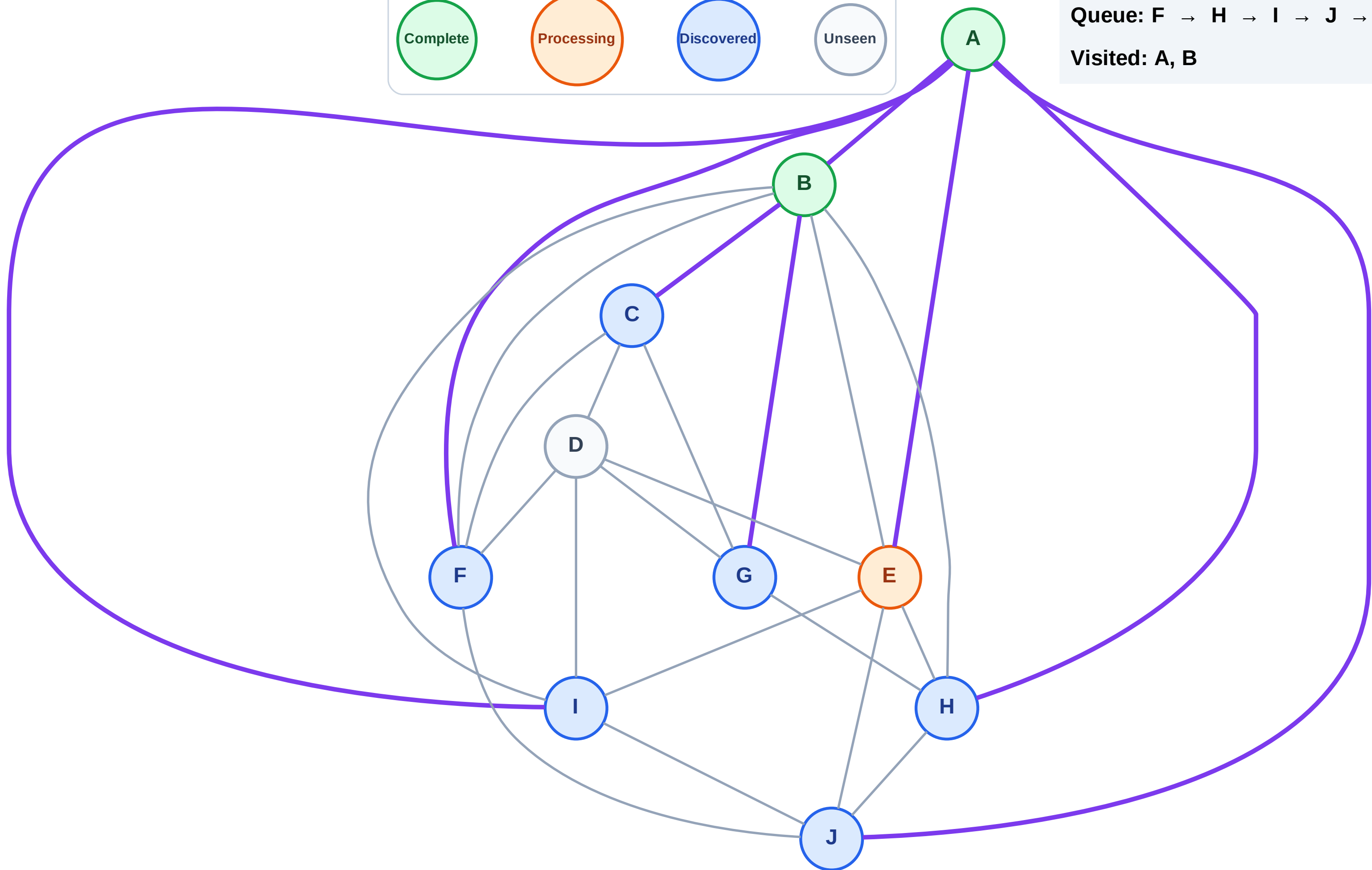
Step 6: Process node E

Legend



Queue: F → H → I → J → C → G

Visited: A, B



BFS Traversal

Step 7: Discover D from E

Legend

Complete

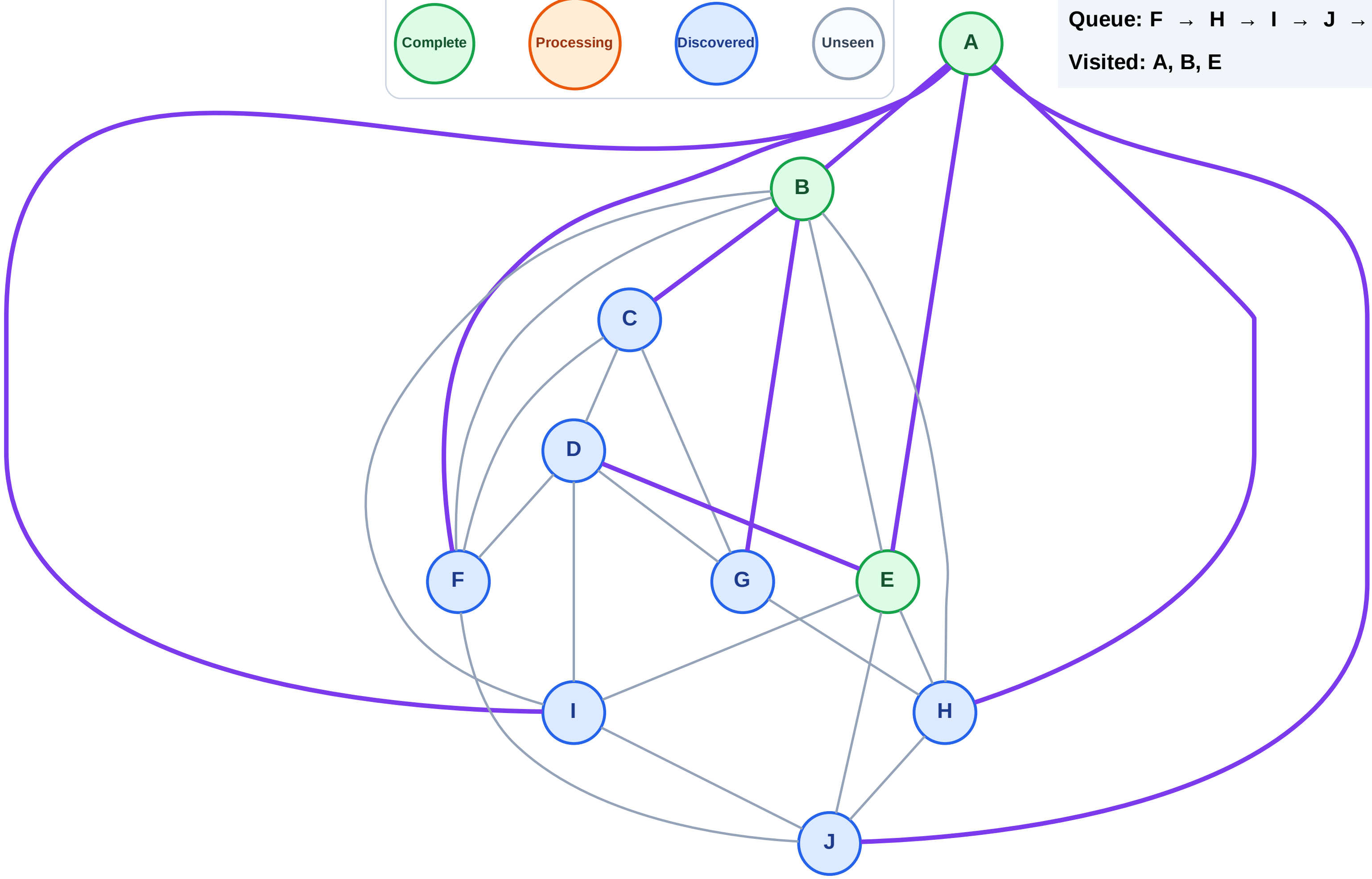
Processing

Discovered

Unseen

Queue: F → H → I → J → C → G → D

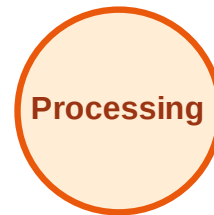
Visited: A, B, E



BFS Traversal

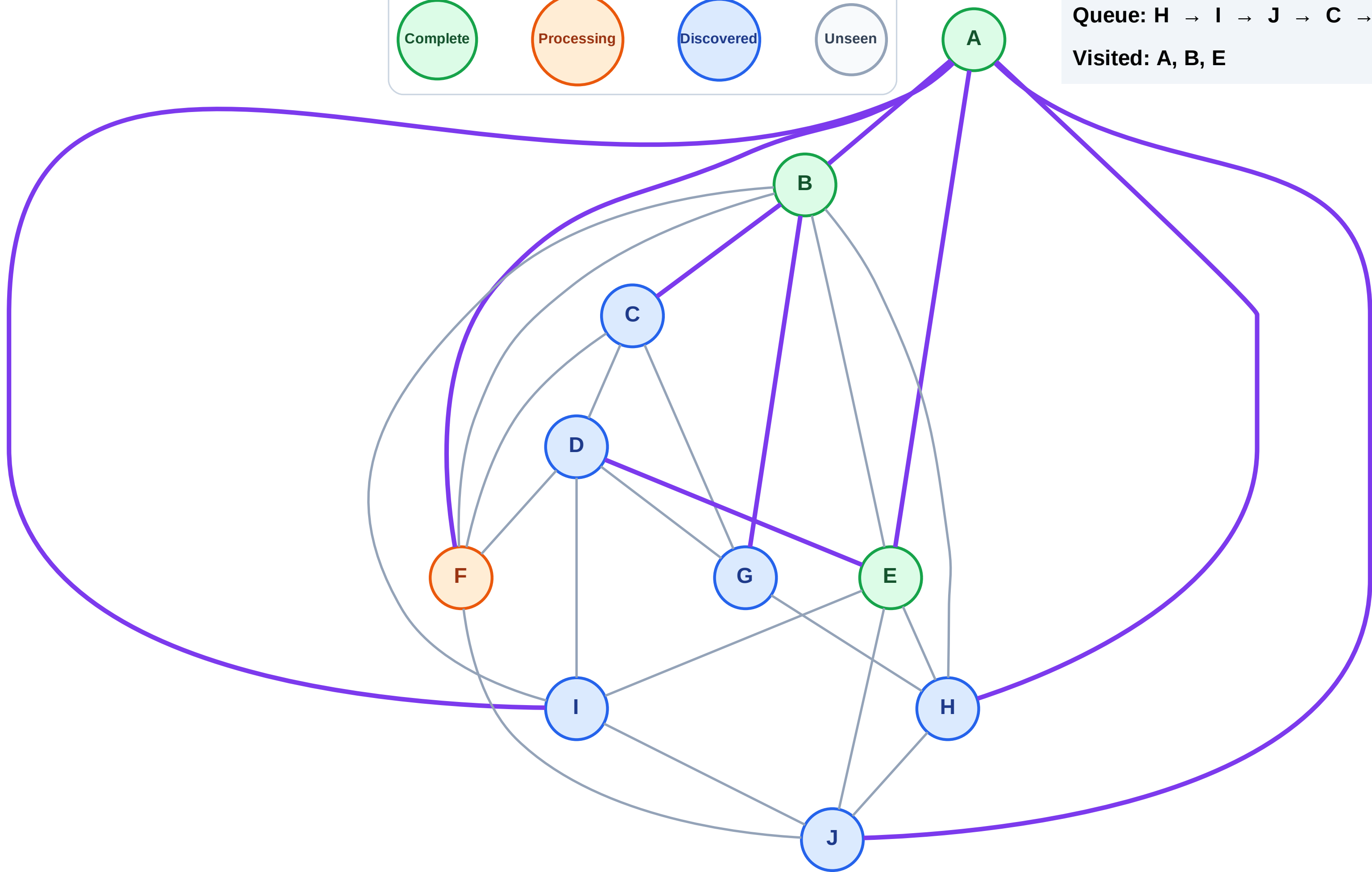
Step 8: Process node F

Legend



Queue: H → I → J → C → G → D

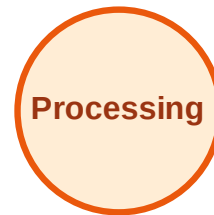
Visited: A, B, E



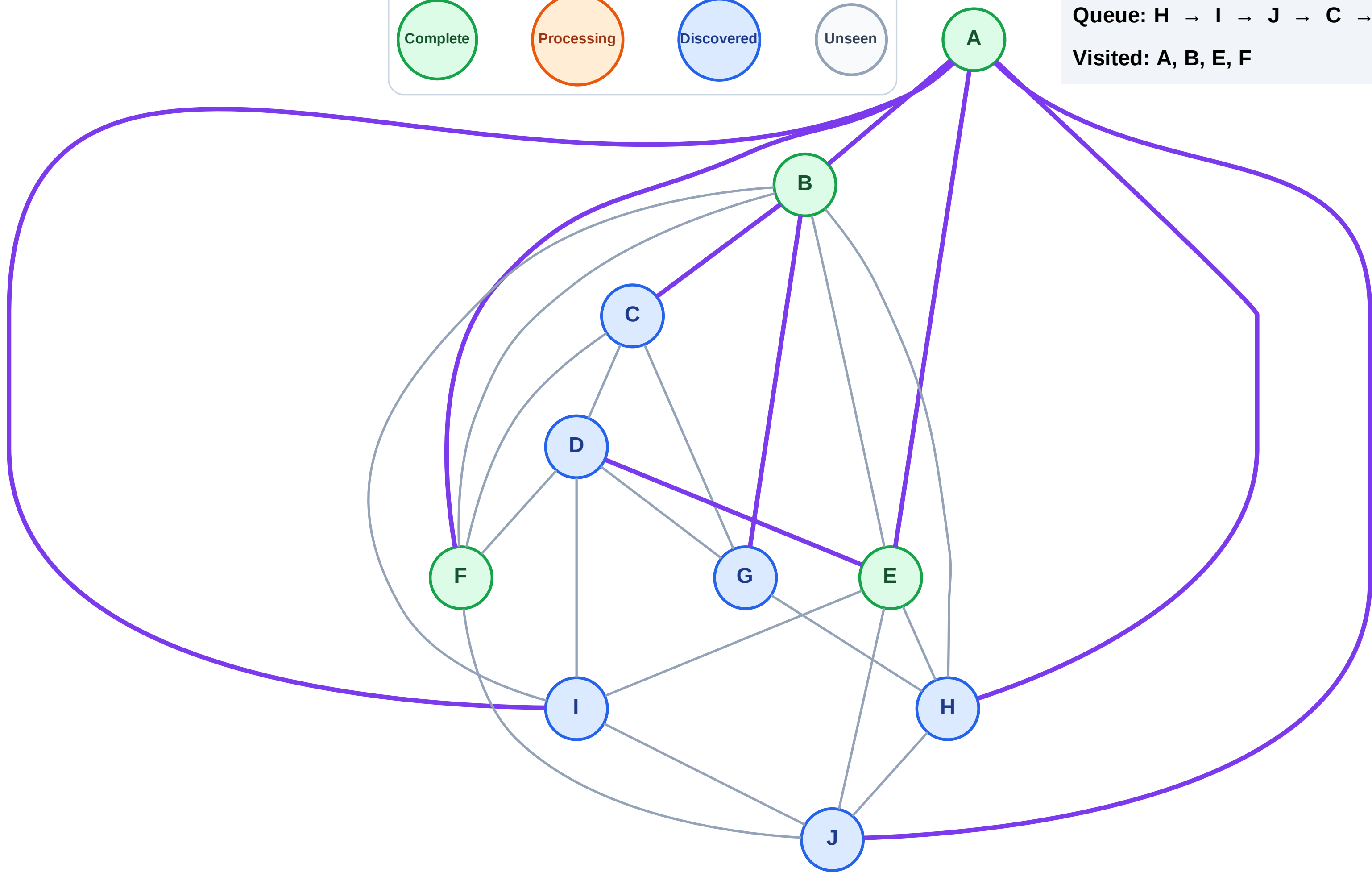
BFS Traversal

Step 9: No new neighbors from F

Legend



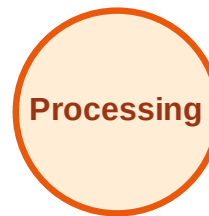
Queue: H → I → J → C → G → D
Visited: A, B, E, F



BFS Traversal

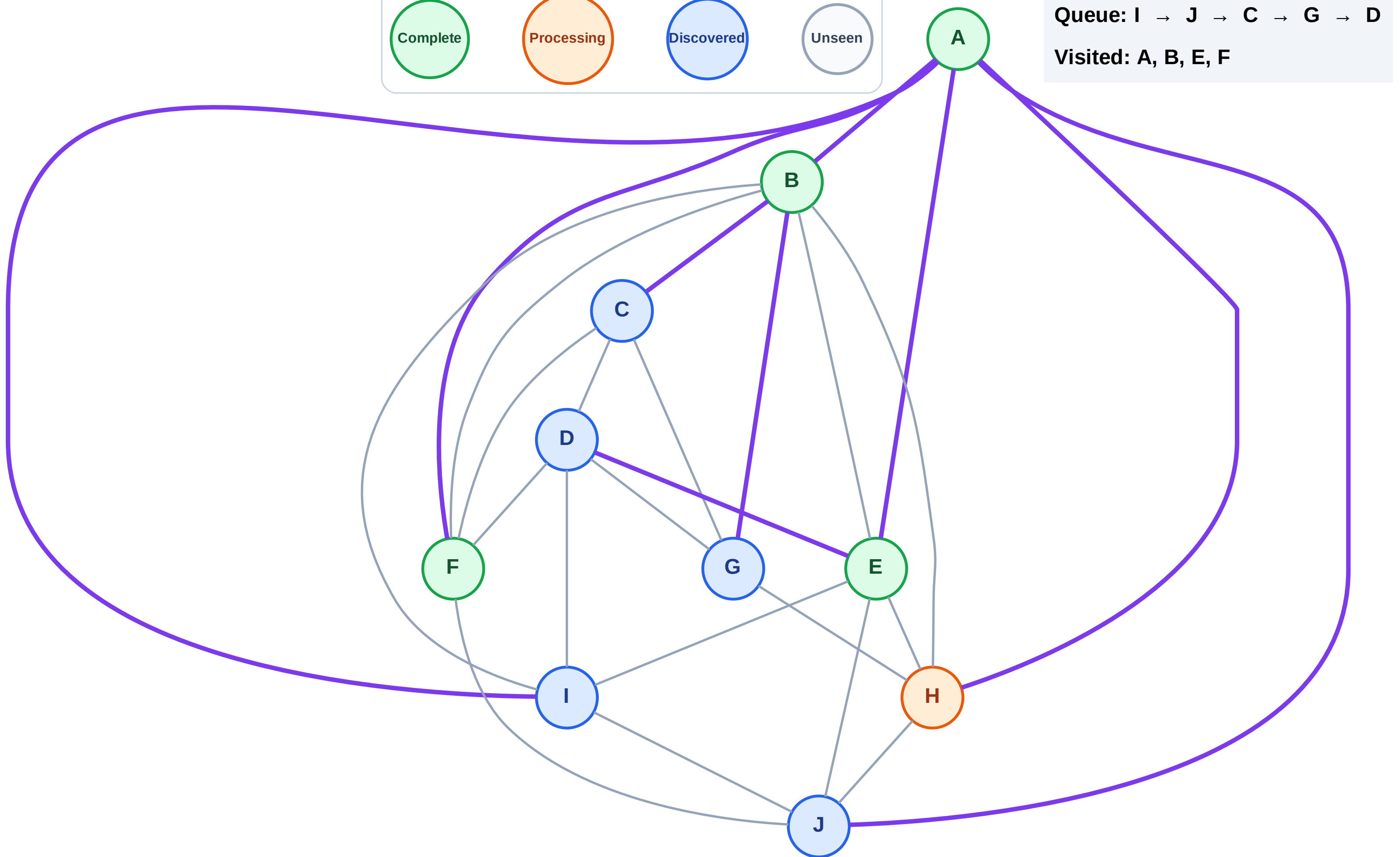
Step 10: Process node H

Legend



Queue: I → J → C → G → D

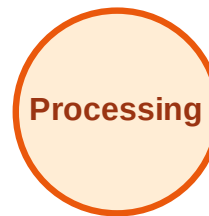
Visited: A, B, E, F



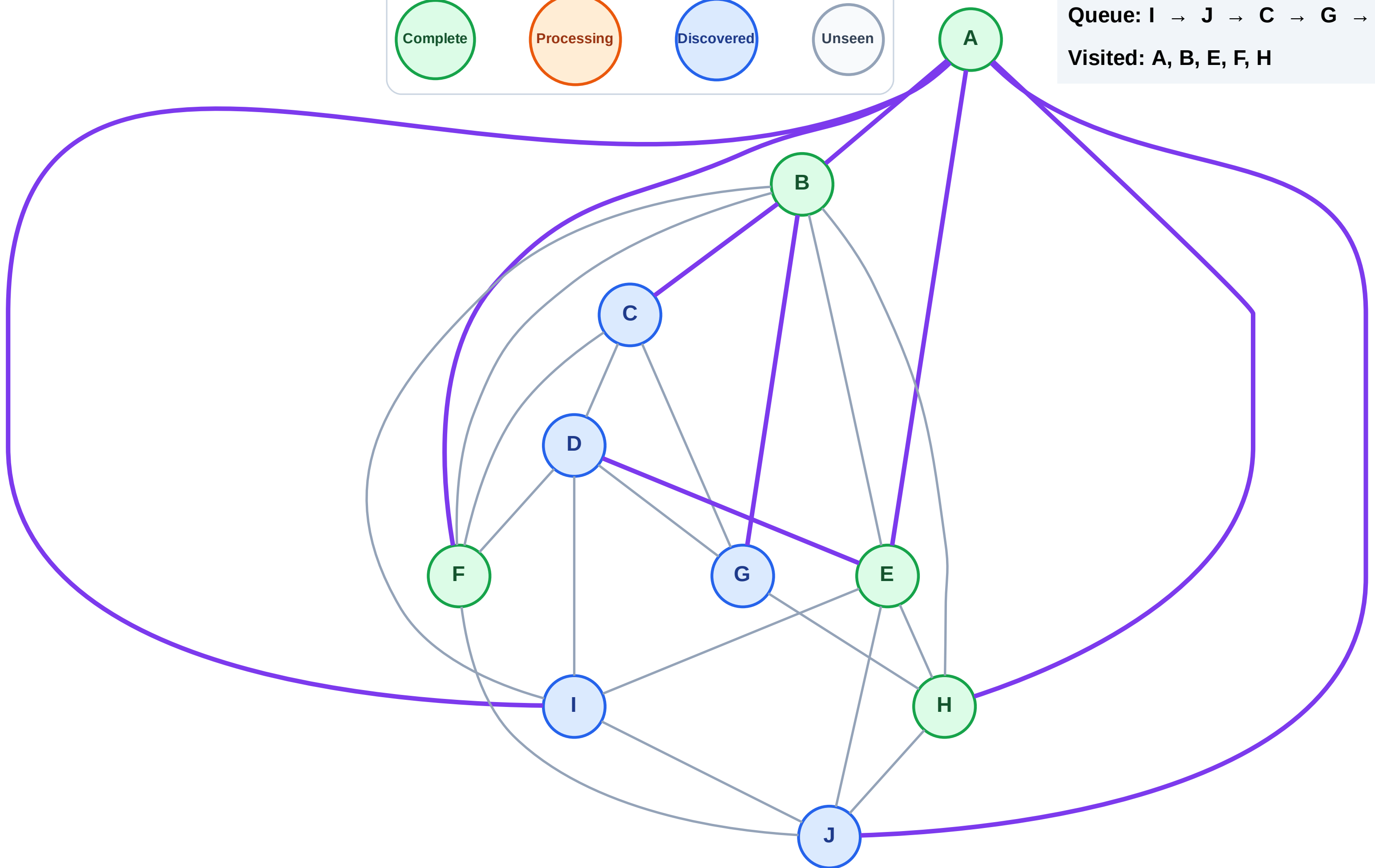
BFS Traversal

Step 11: No new neighbors from H

Legend



Queue: I → J → C → G → D
Visited: A, B, E, F, H



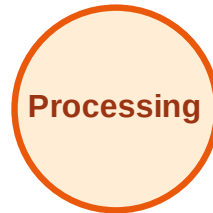
BFS Traversal

Step 12: Process node I

Legend



Complete



Processing



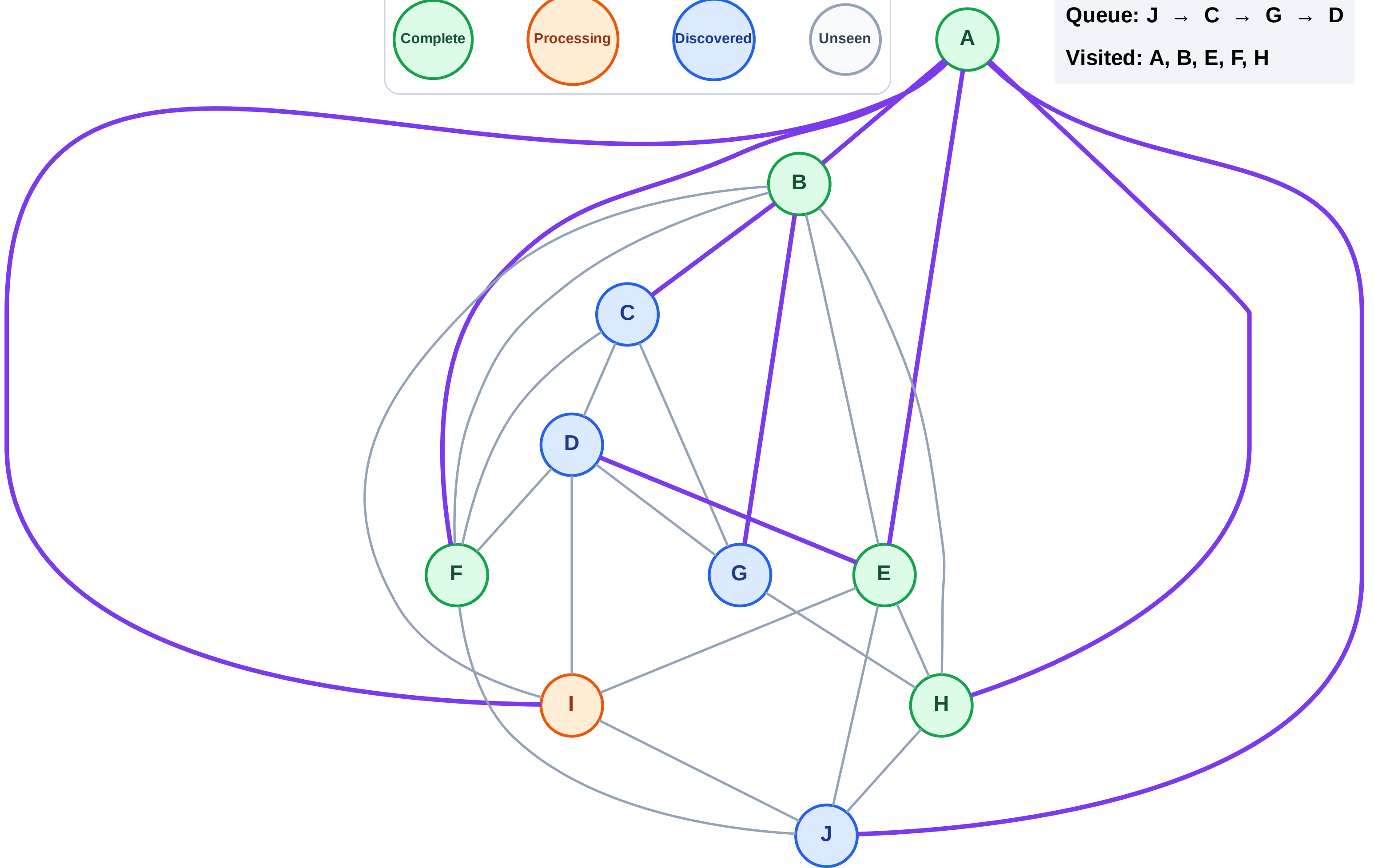
Discovered



Unseen

Queue: J → C → G → D

Visited: A, B, E, F, H



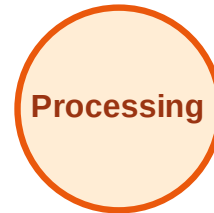
BFS Traversal

Step 13: No new neighbors from I

Legend



Complete



Processing

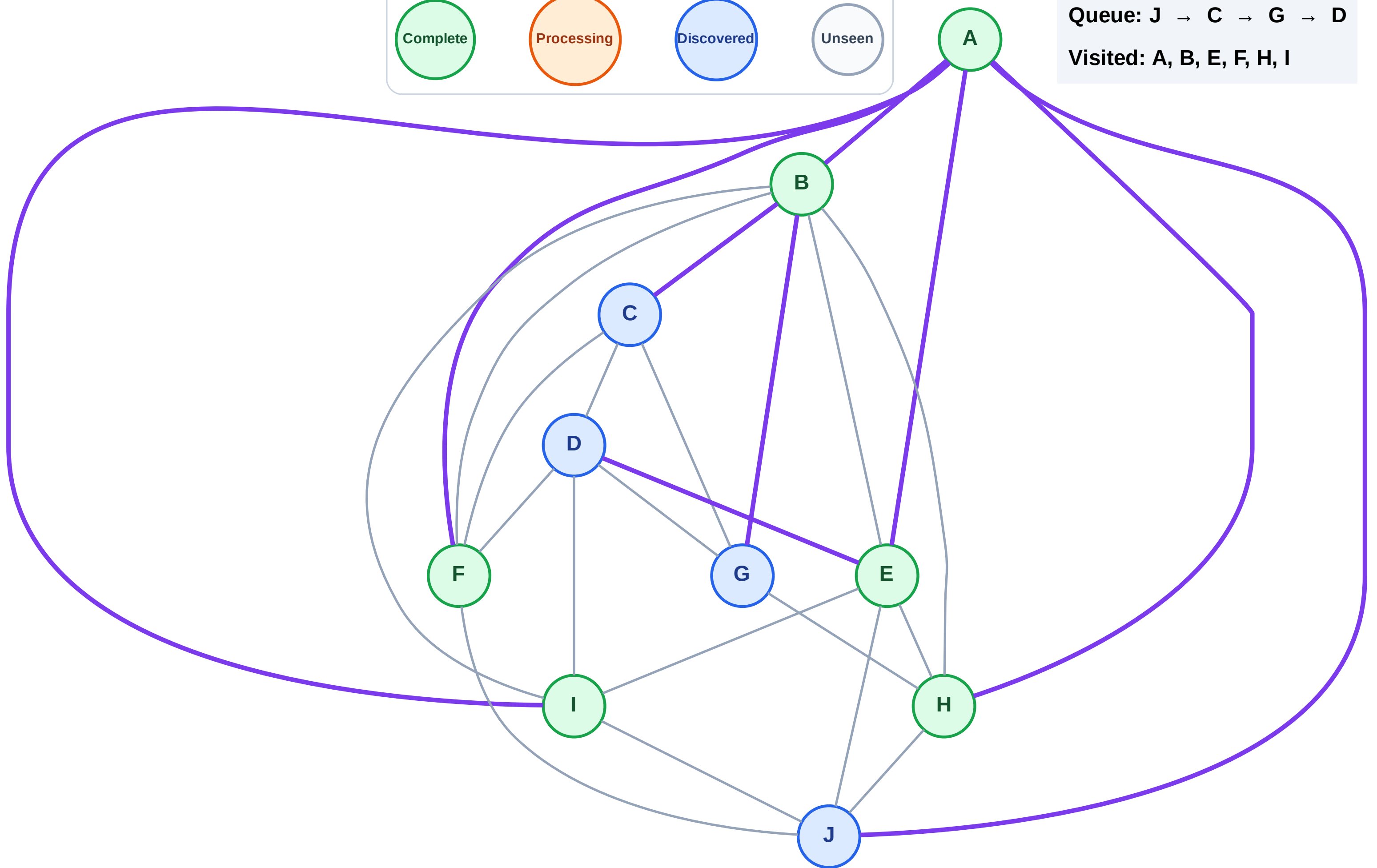


Discovered



Unseen

Queue: J → C → G → D
Visited: A, B, E, F, H, I



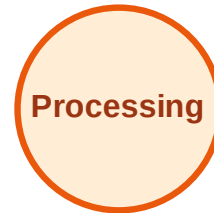
BFS Traversal

Step 14: Process node J

Legend



Complete



Processing



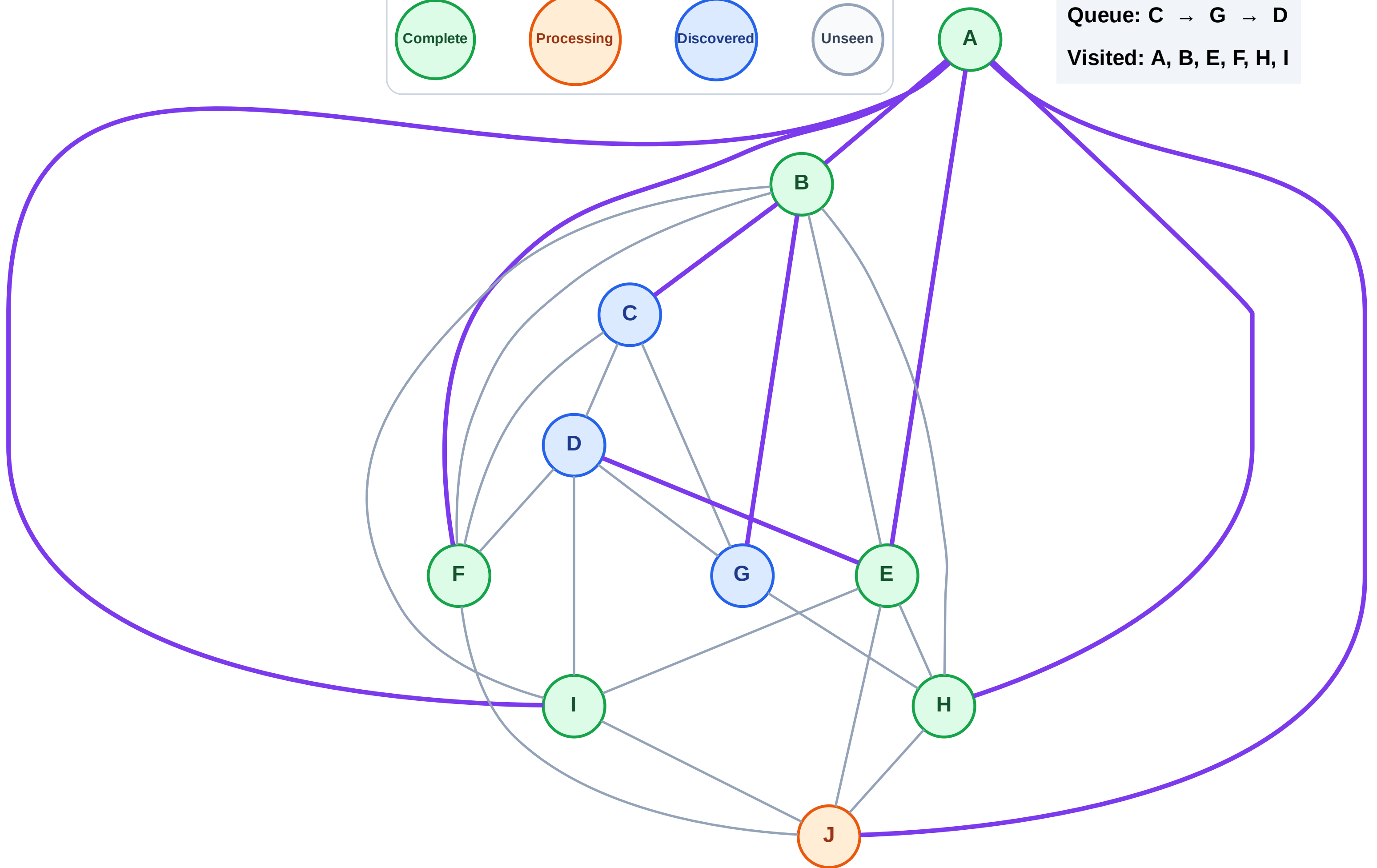
Discovered



Unseen

Queue: C → G → D

Visited: A, B, E, F, H, I



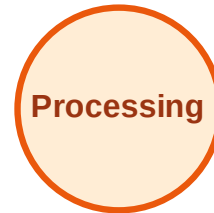
BFS Traversal

Step 15: No new neighbors from J

Legend



Complete



Processing



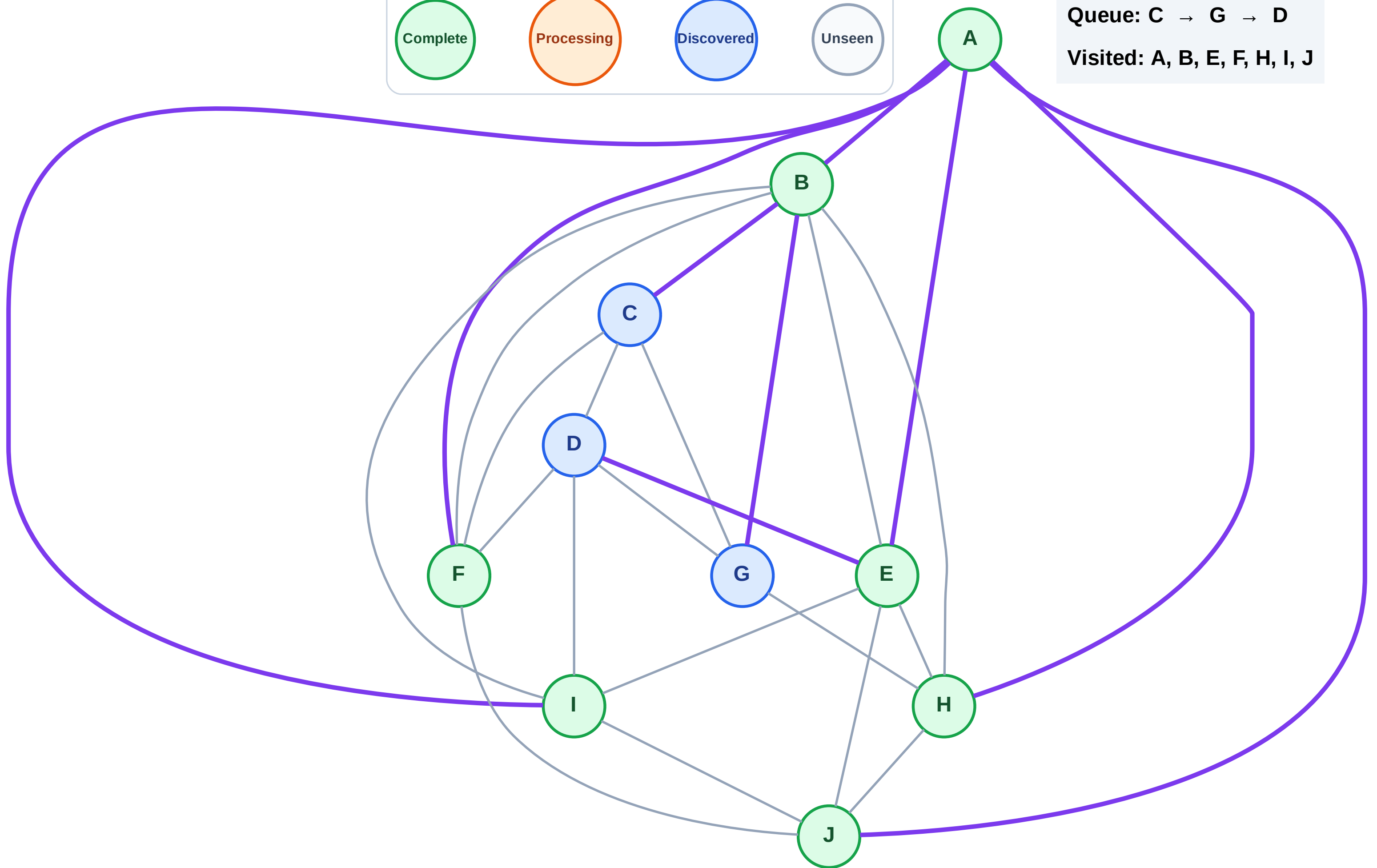
Discovered



Unseen

Queue: C → G → D

Visited: A, B, E, F, H, I, J



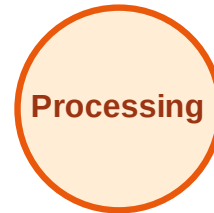
BFS Traversal

Step 16: Process node C

Legend



Complete



Processing



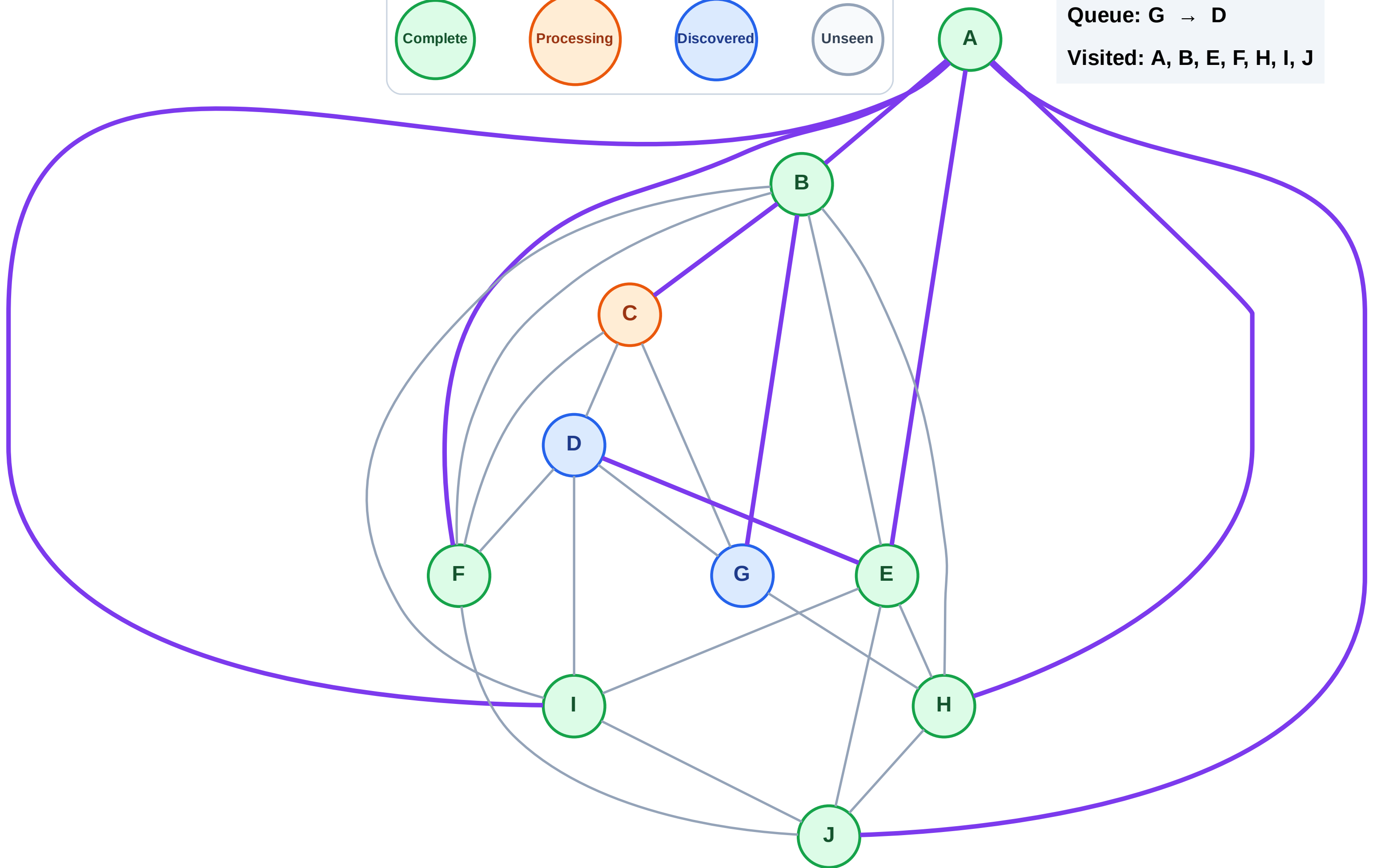
Discovered



Unseen

Queue: G → D

Visited: A, B, E, F, H, I, J



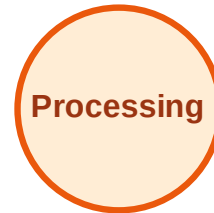
BFS Traversal

Step 17: No new neighbors from C

Legend



Complete



Processing



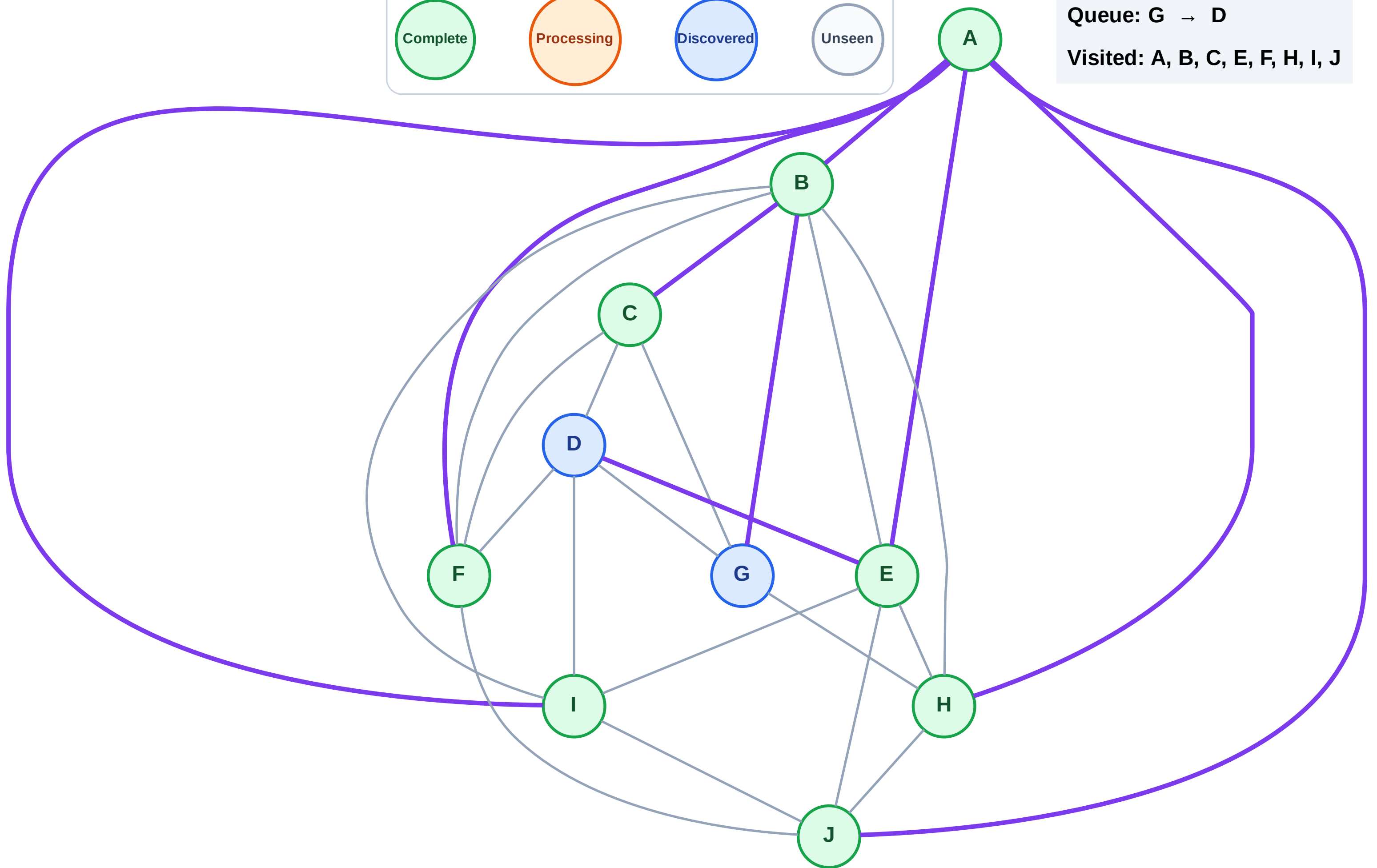
Discovered



Unseen

Queue: G → D

Visited: A, B, C, E, F, H, I, J



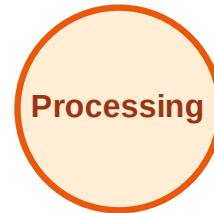
BFS Traversal

Step 18: Process node G

Legend



Complete



Processing



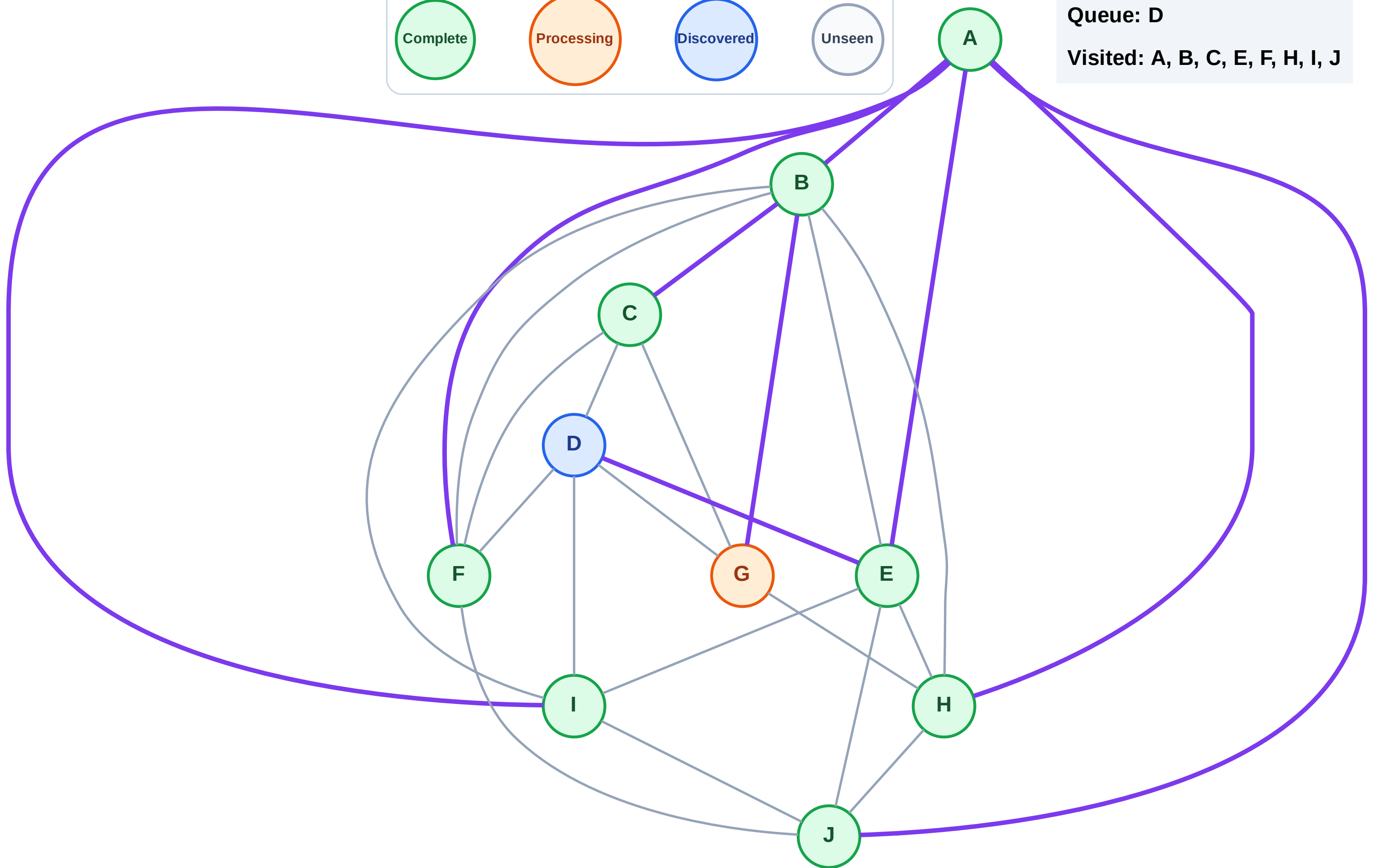
Discovered



Unseen

Queue: D

Visited: A, B, C, E, F, H, I, J



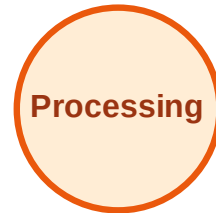
BFS Traversal

Step 19: No new neighbors from G

Legend



Complete



Processing



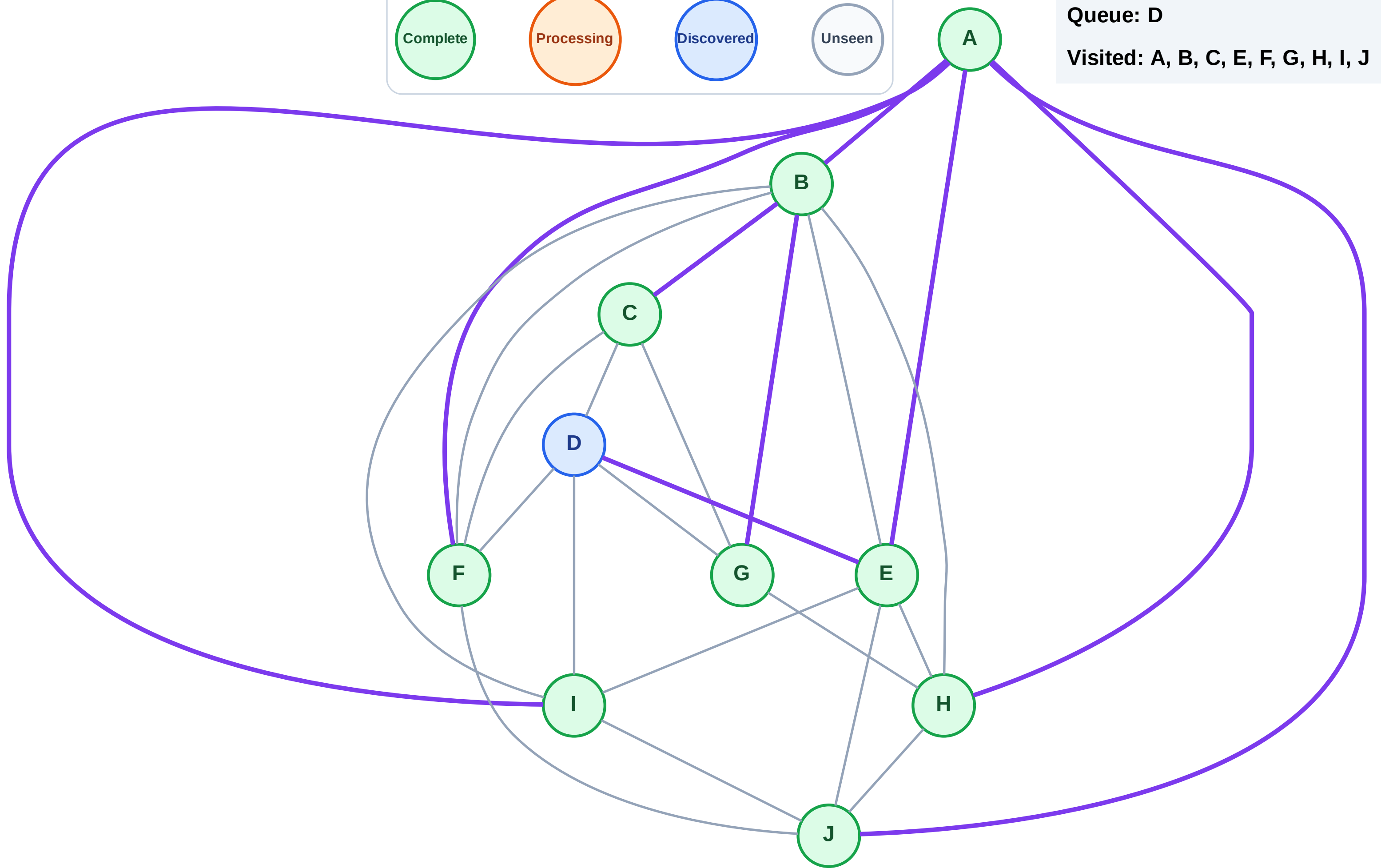
Discovered



Unseen

Queue: D

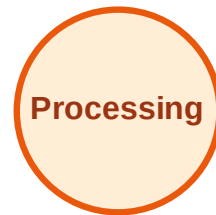
Visited: A, B, C, E, F, G, H, I, J



BFS Traversal

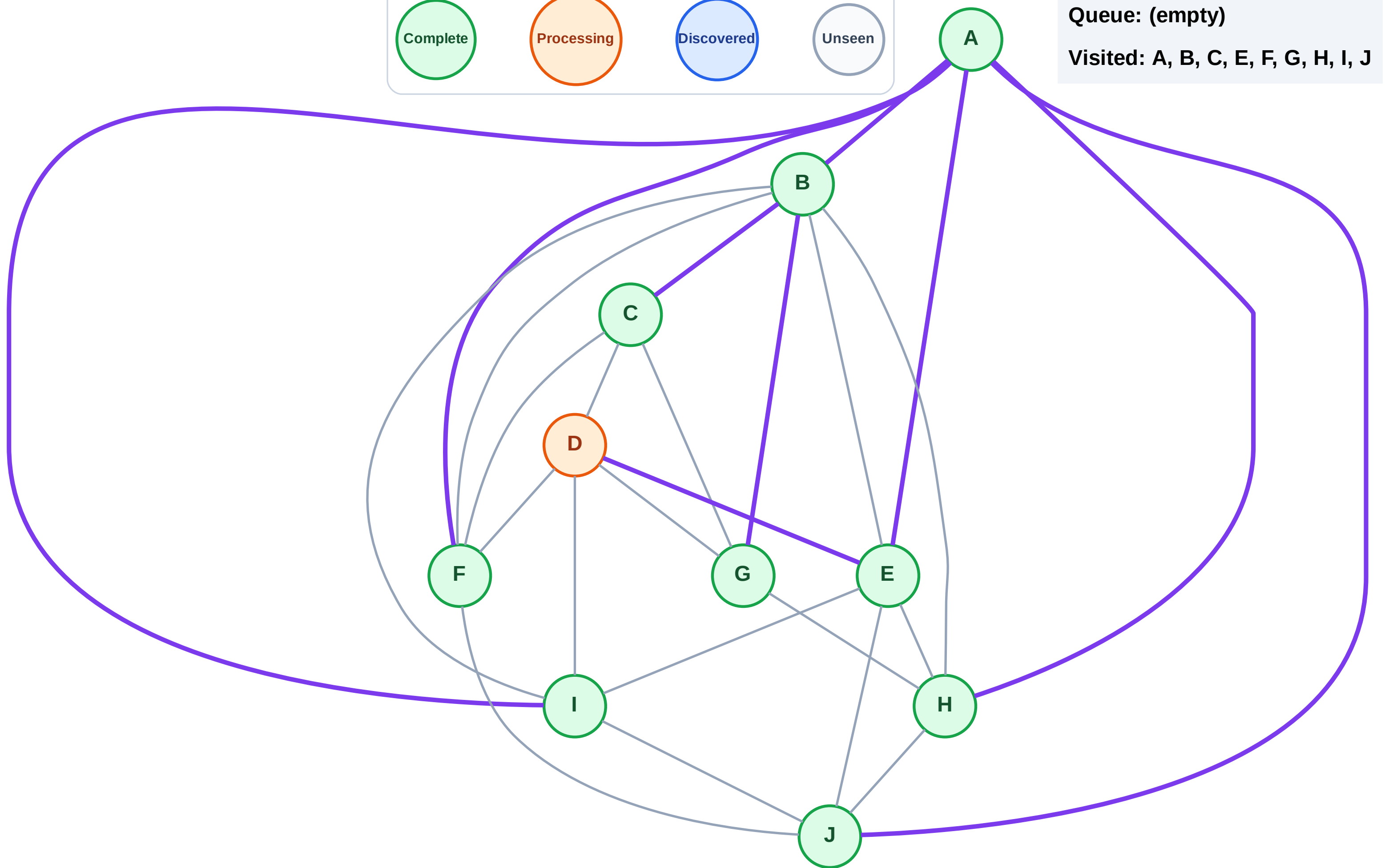
Step 20: Process node D

Legend



Queue: (empty)

Visited: A, B, C, E, F, G, H, I, J



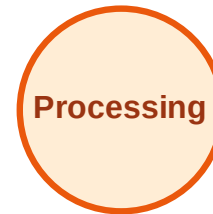
BFS Traversal

Step 21: No new neighbors from D

Legend



Complete



Processing



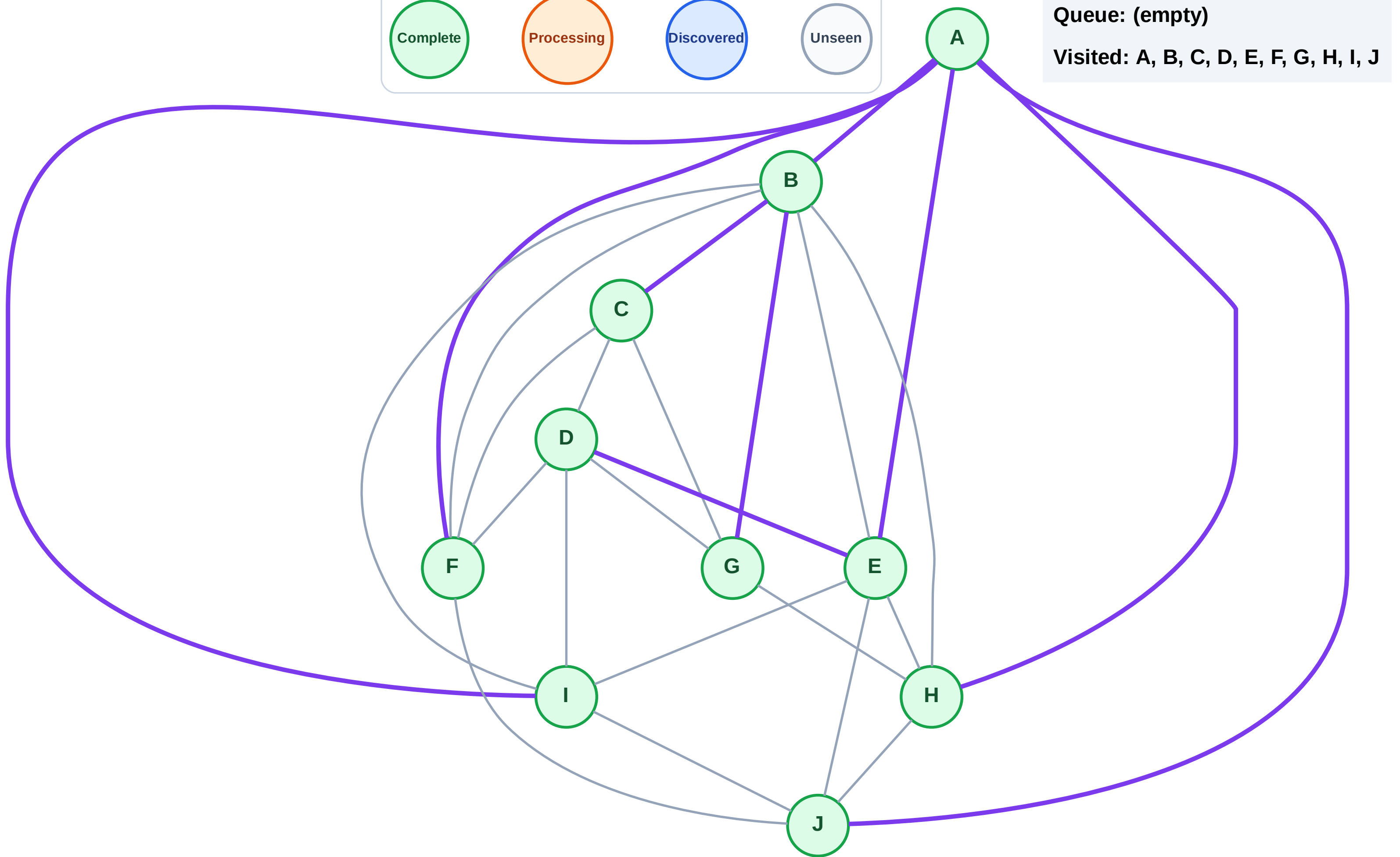
Discovered



Unseen

Queue: (empty)

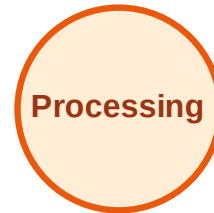
Visited: A, B, C, D, E, F, G, H, I, J



DFS Traversal

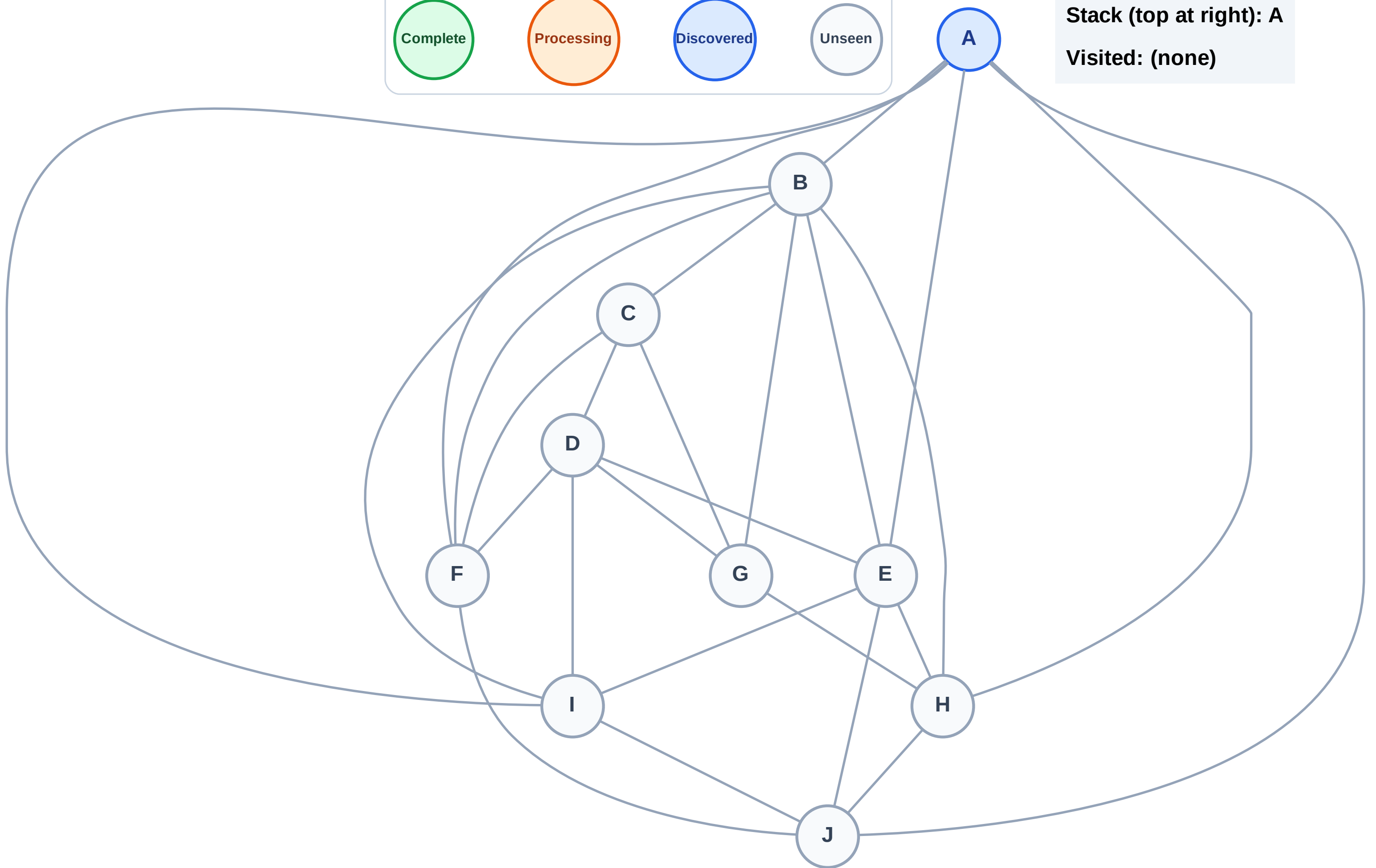
Step 1: Start at A

Legend



Stack (top at right): A

Visited: (none)



DFS Traversal

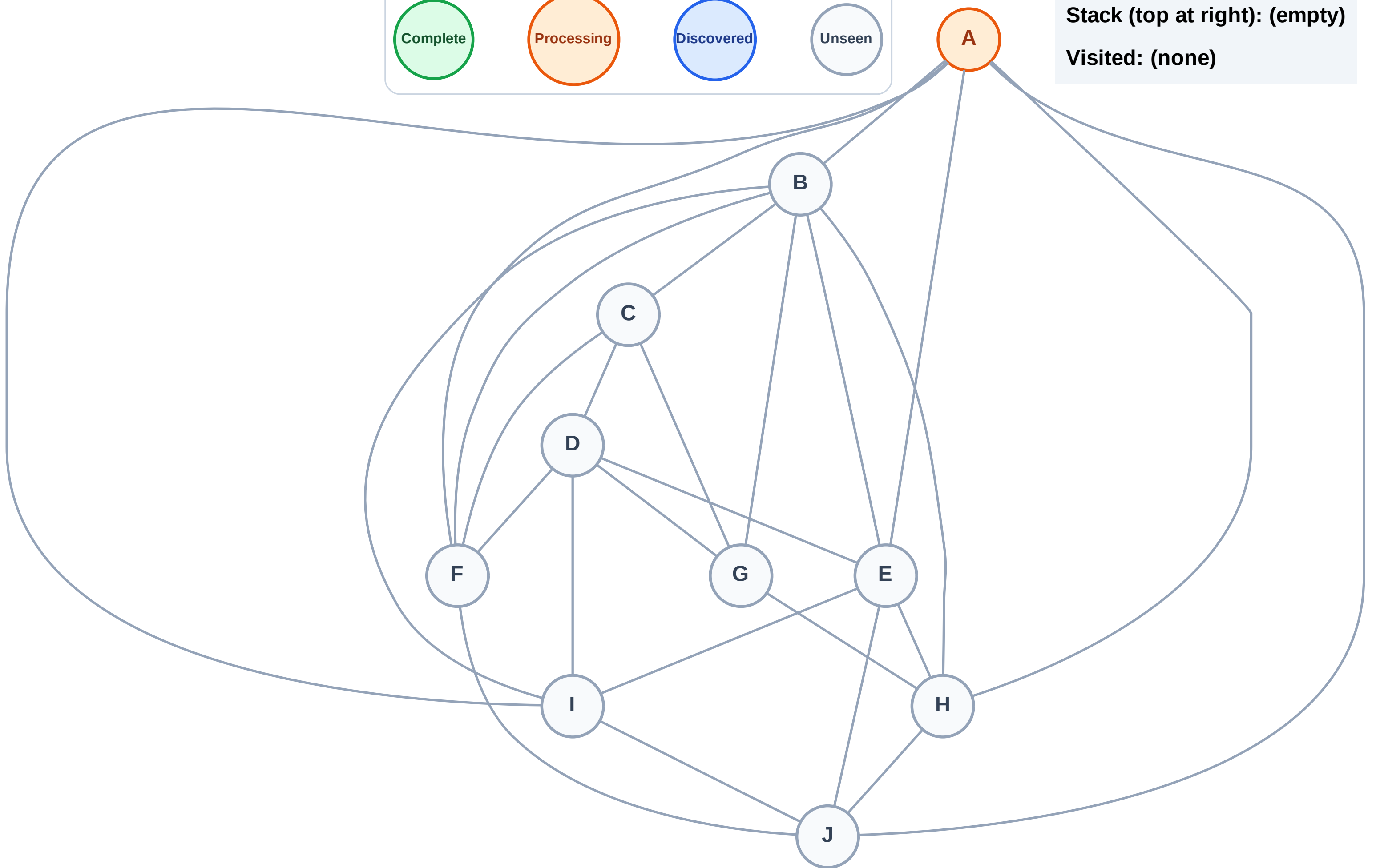
Step 2: Process node A

Legend



Stack (top at right): (empty)

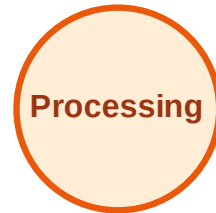
Visited: (none)



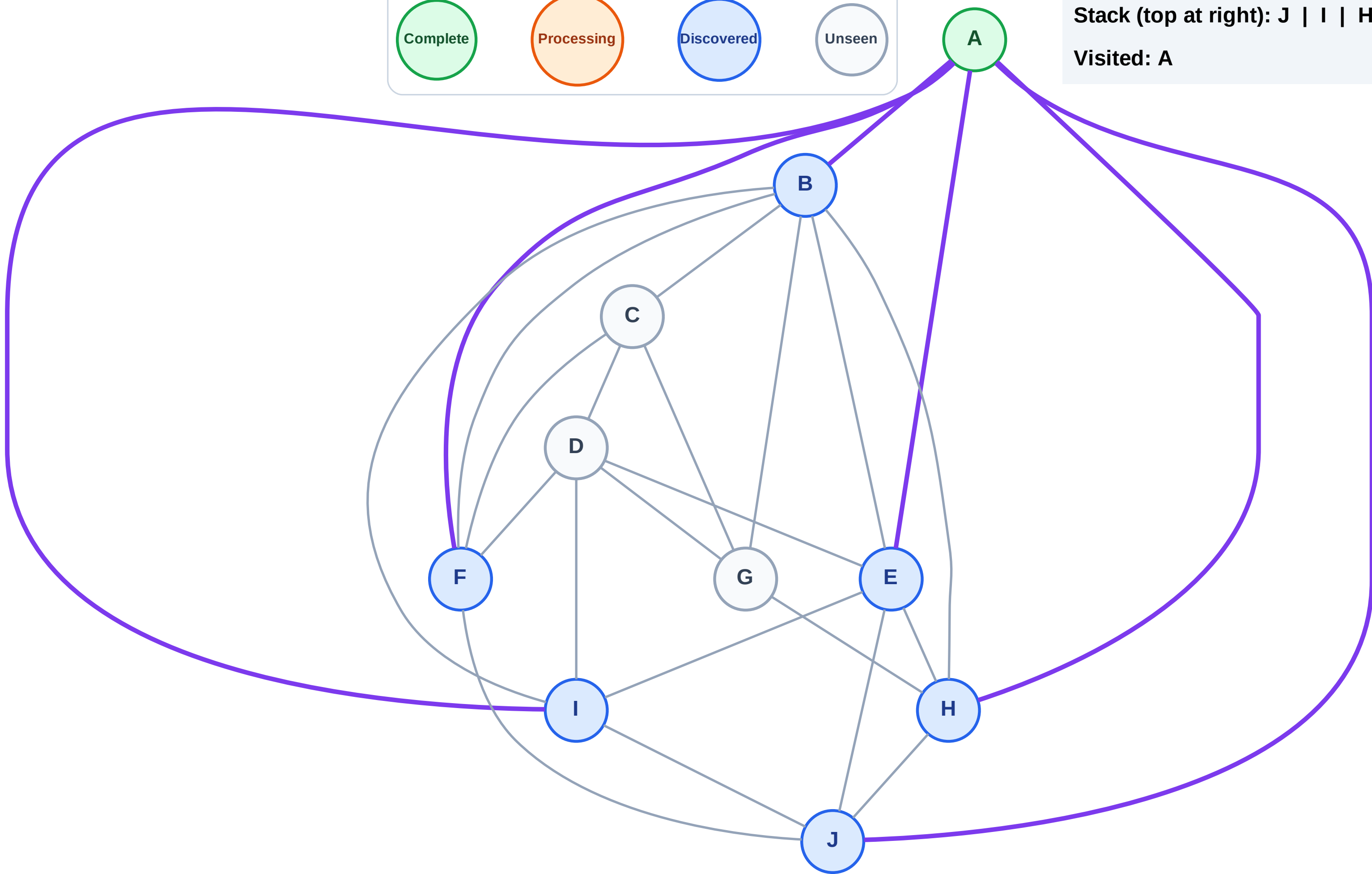
DFS Traversal

Step 3: Discover J, I, H, F, E, B from A

Legend



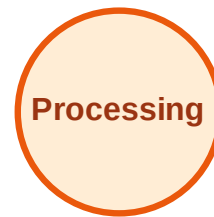
Stack (top at right): J | I | H | F | E | B
Visited: A



DFS Traversal

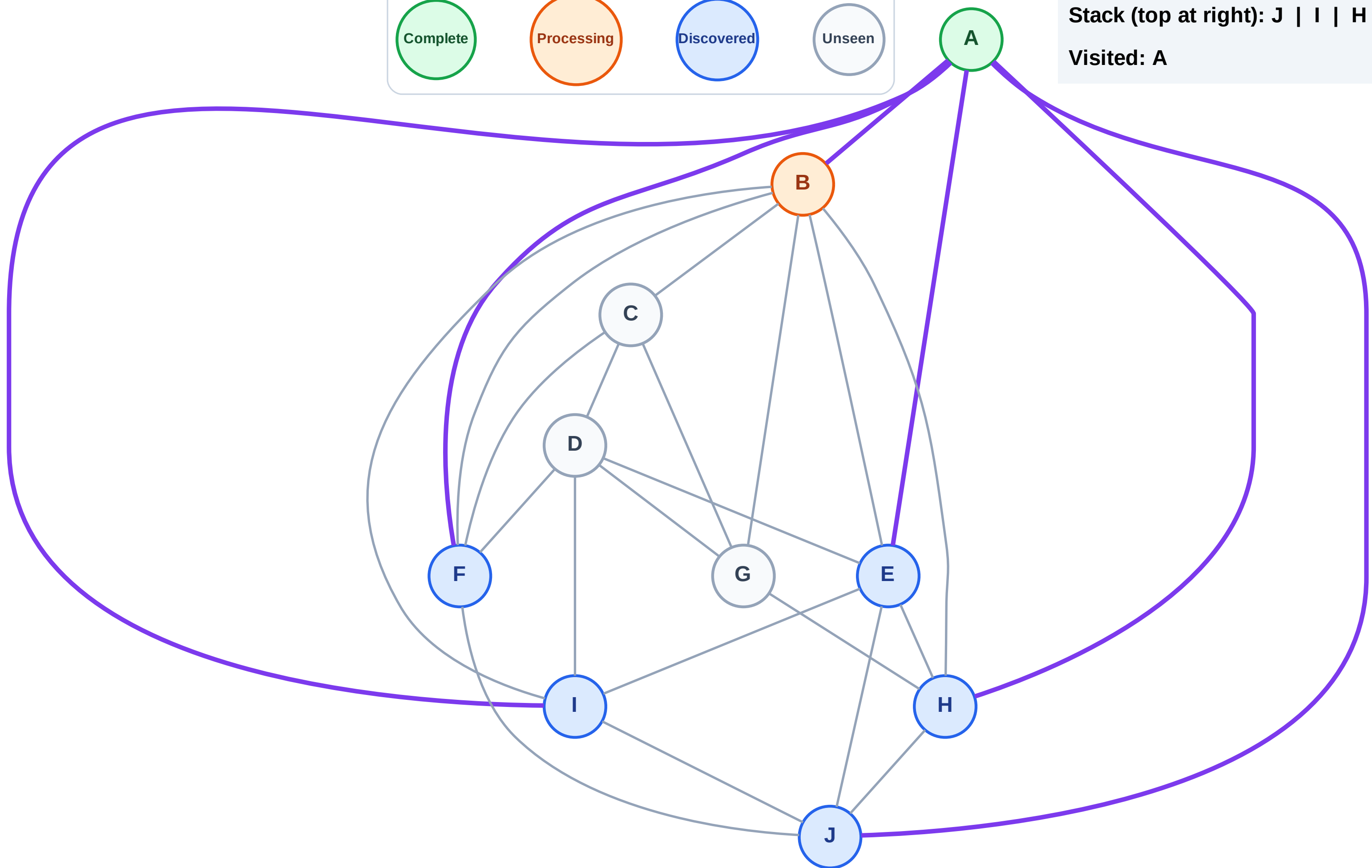
Step 4: Process node B

Legend



Stack (top at right): J | I | H | F | E

Visited: A



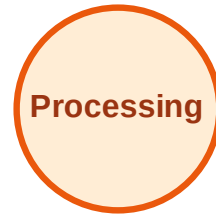
DFS Traversal

Step 5: Discover G, C from B

Legend



Complete



Processing



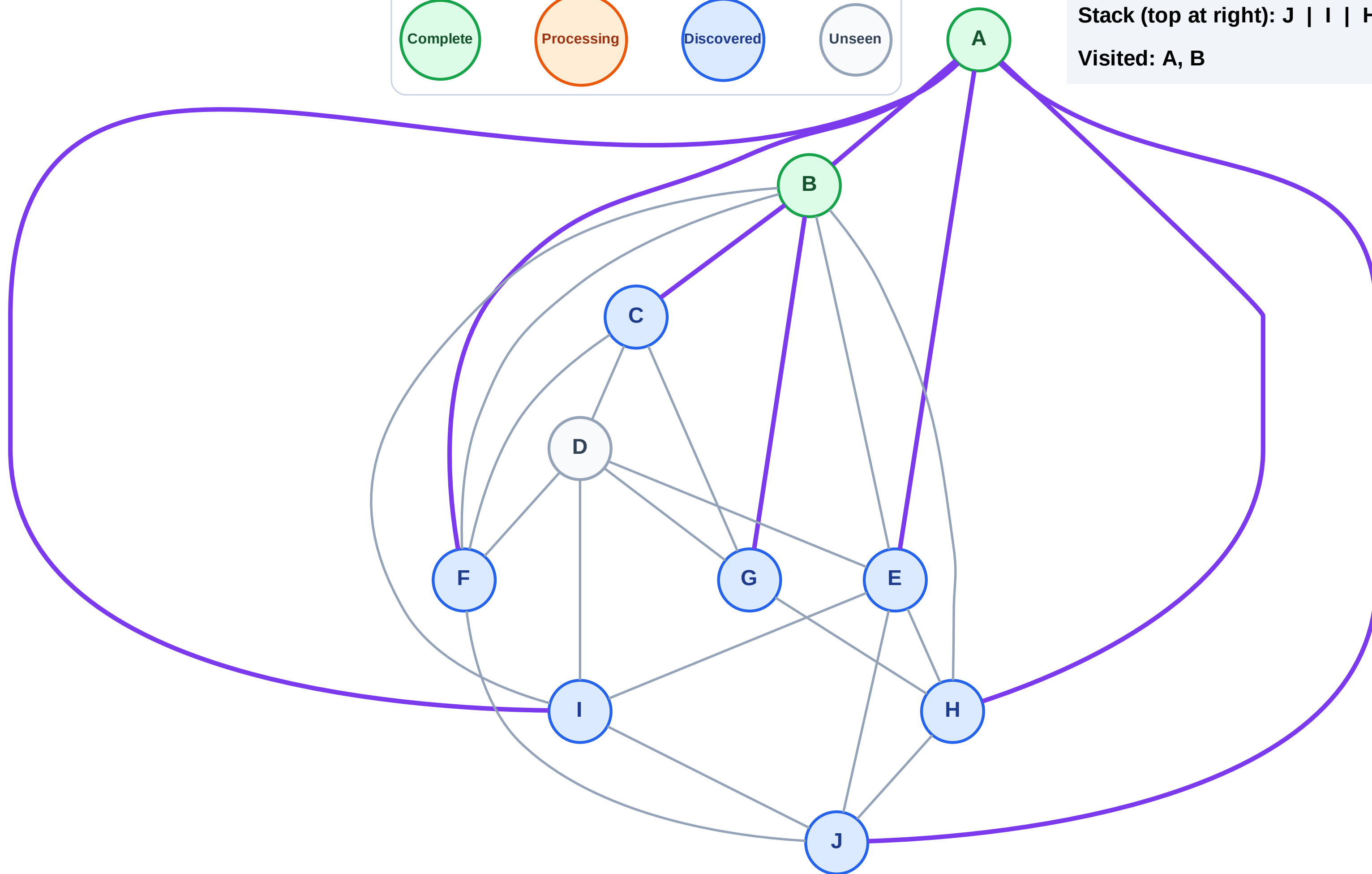
Discovered



Unseen

Stack (top at right): J | I | H | F | E | G | C

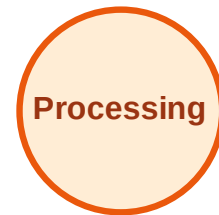
Visited: A, B



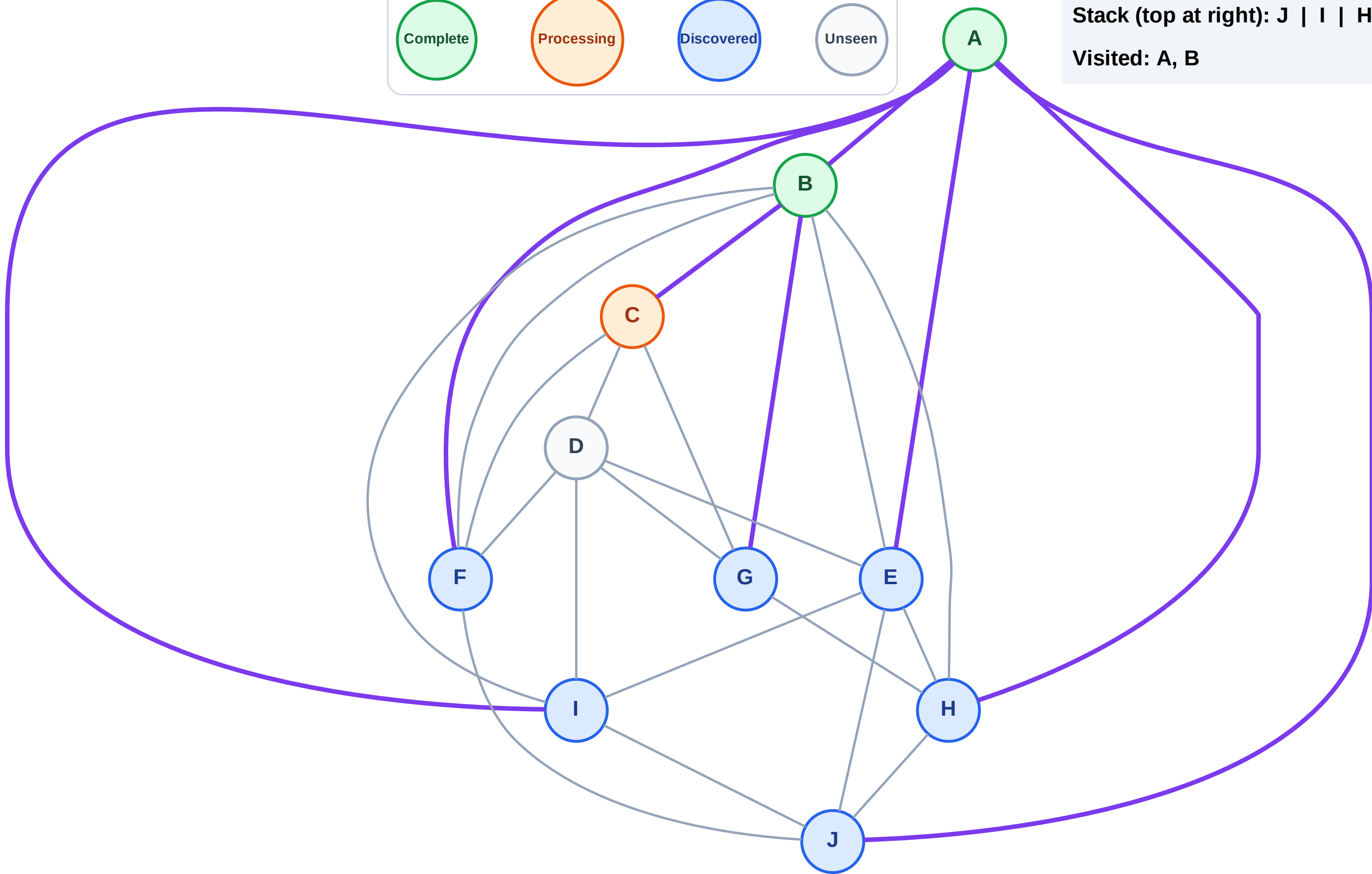
DFS Traversal

Step 6: Process node C

Legend



Stack (top at right): J | I | H | F | E | G
Visited: A, B



DFS Traversal

Step 7: Discover D from C

Legend

Complete

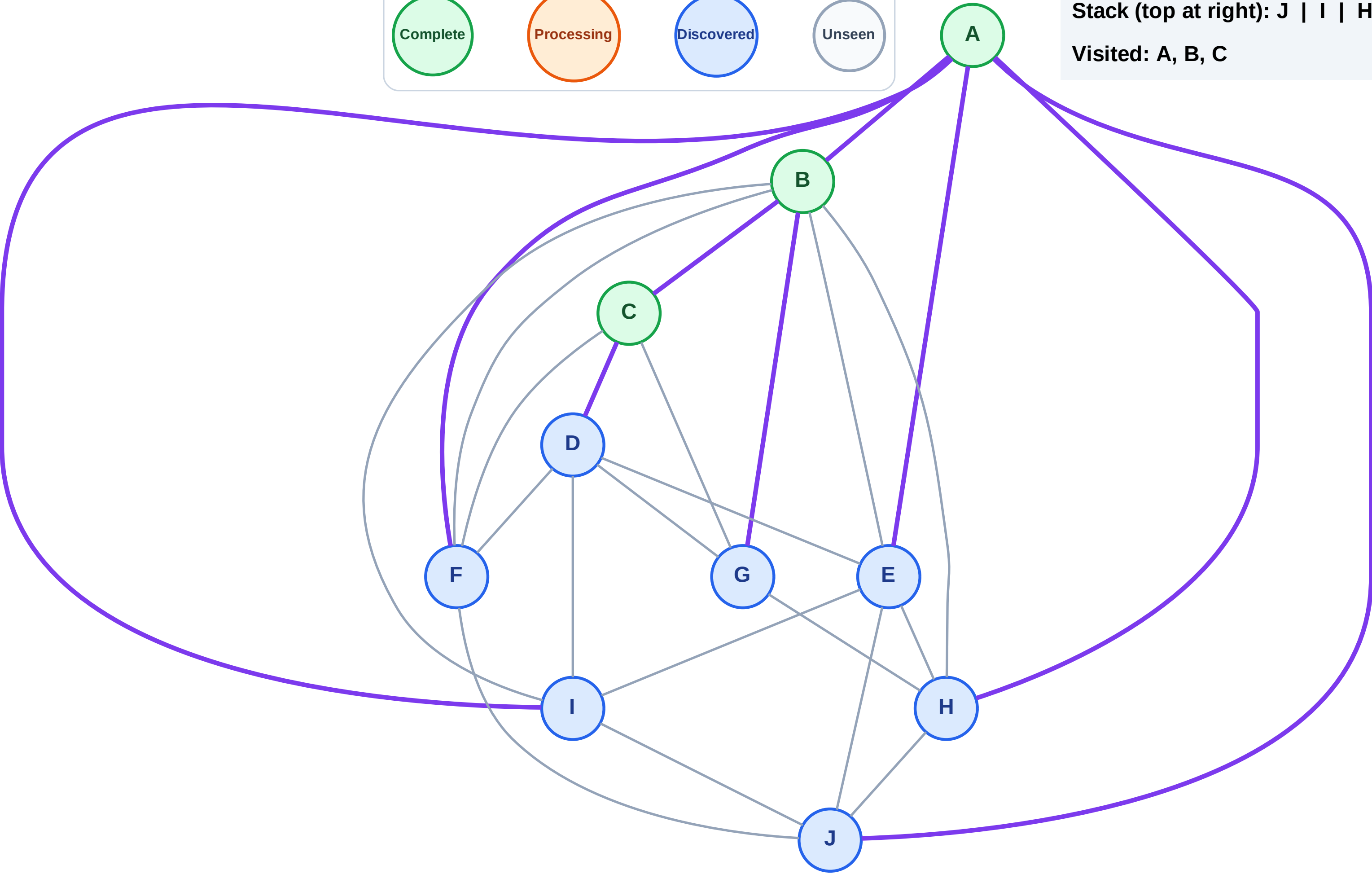
Processing

Discovered

Unseen

Stack (top at right): J | I | H | F | E | G | D

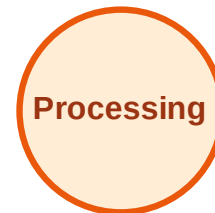
Visited: A, B, C



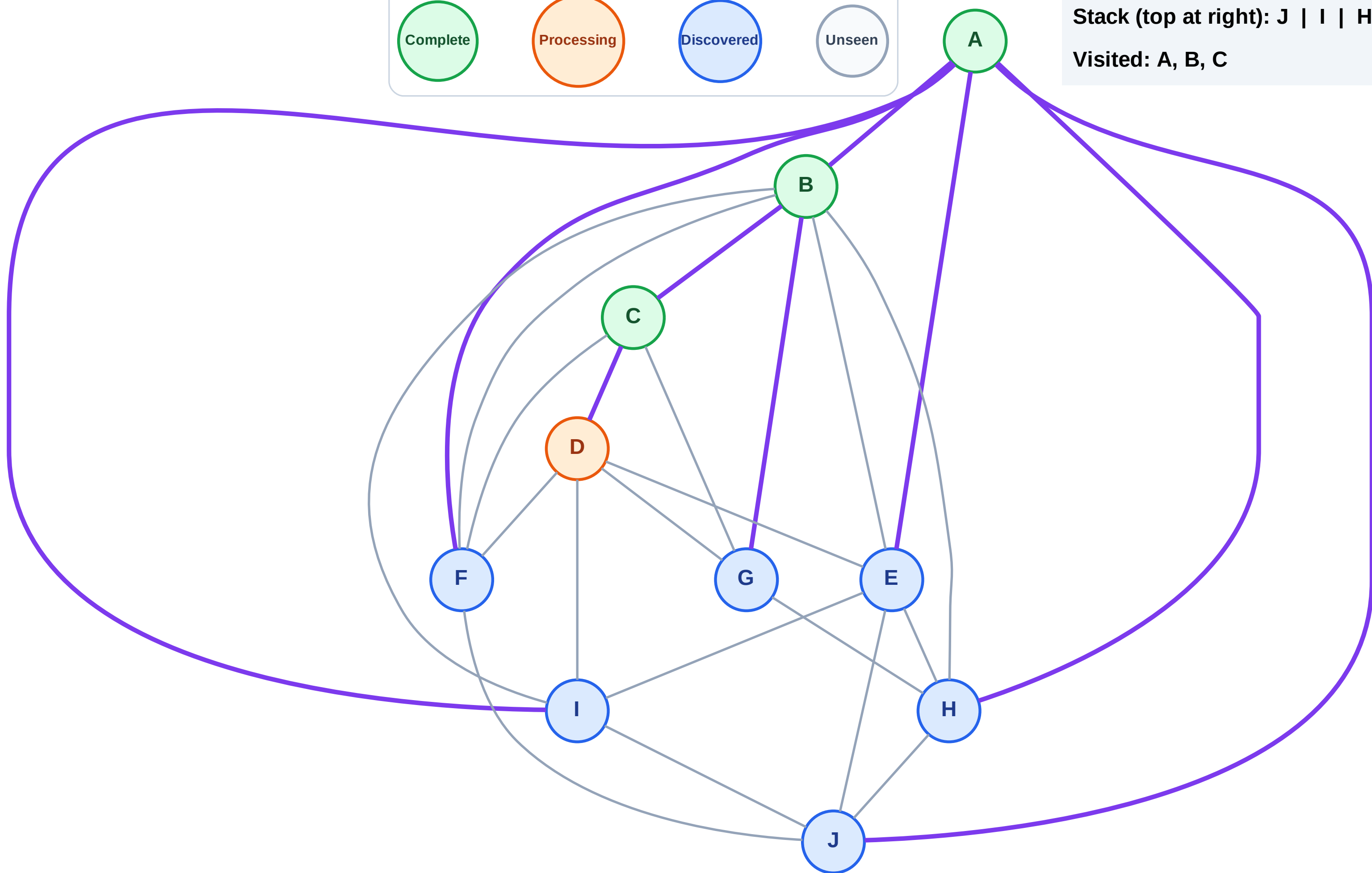
DFS Traversal

Step 8: Process node D

Legend



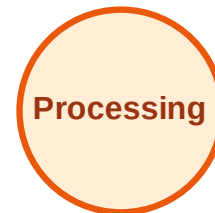
Stack (top at right): J | I | H | F | E | G
Visited: A, B, C



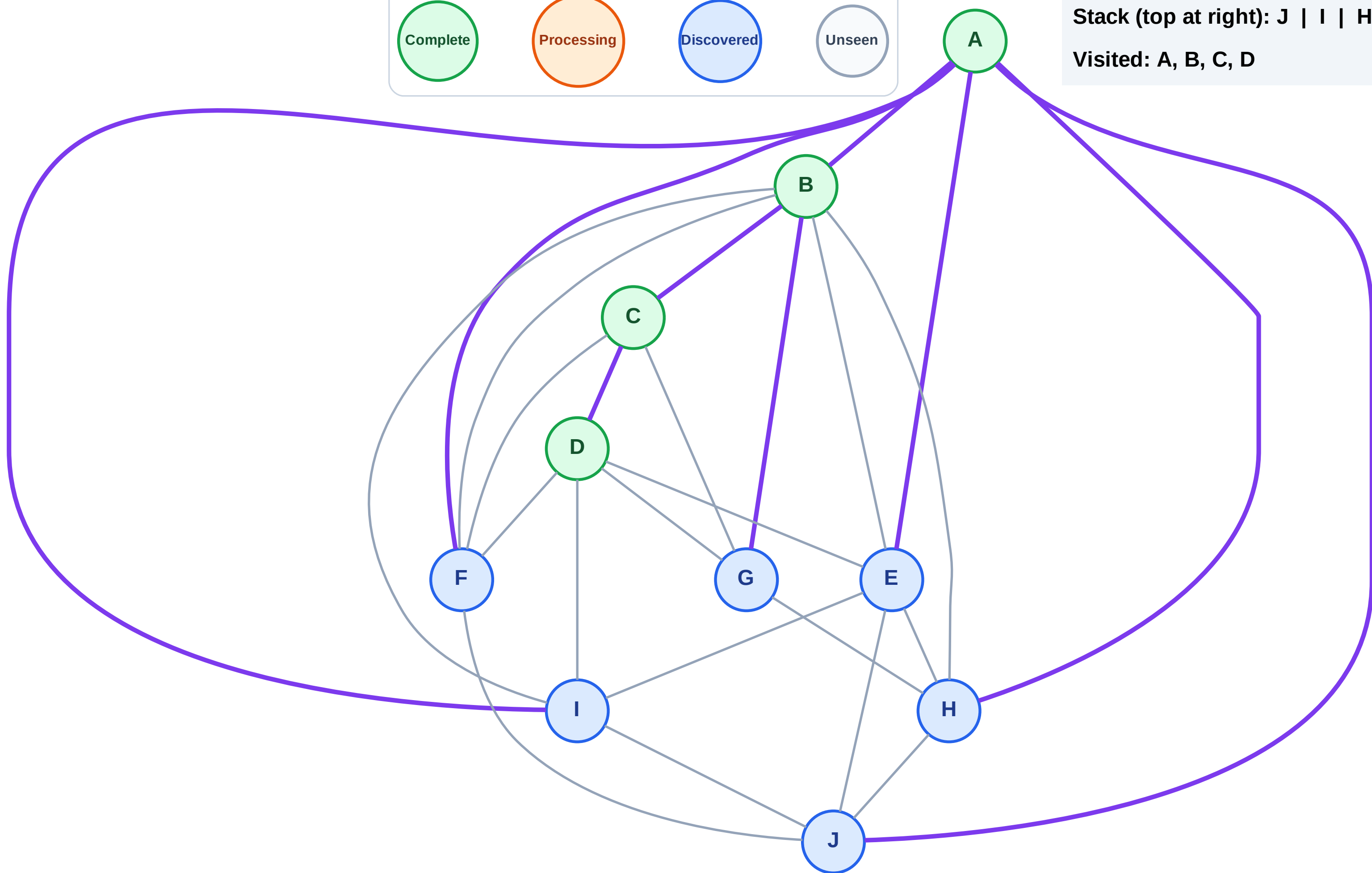
DFS Traversal

Step 9: No new neighbors from D

Legend



Stack (top at right): J | I | H | F | E | G
Visited: A, B, C, D



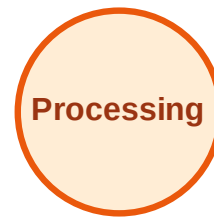
DFS Traversal

Step 10: Process node G

Legend



Complete



Processing



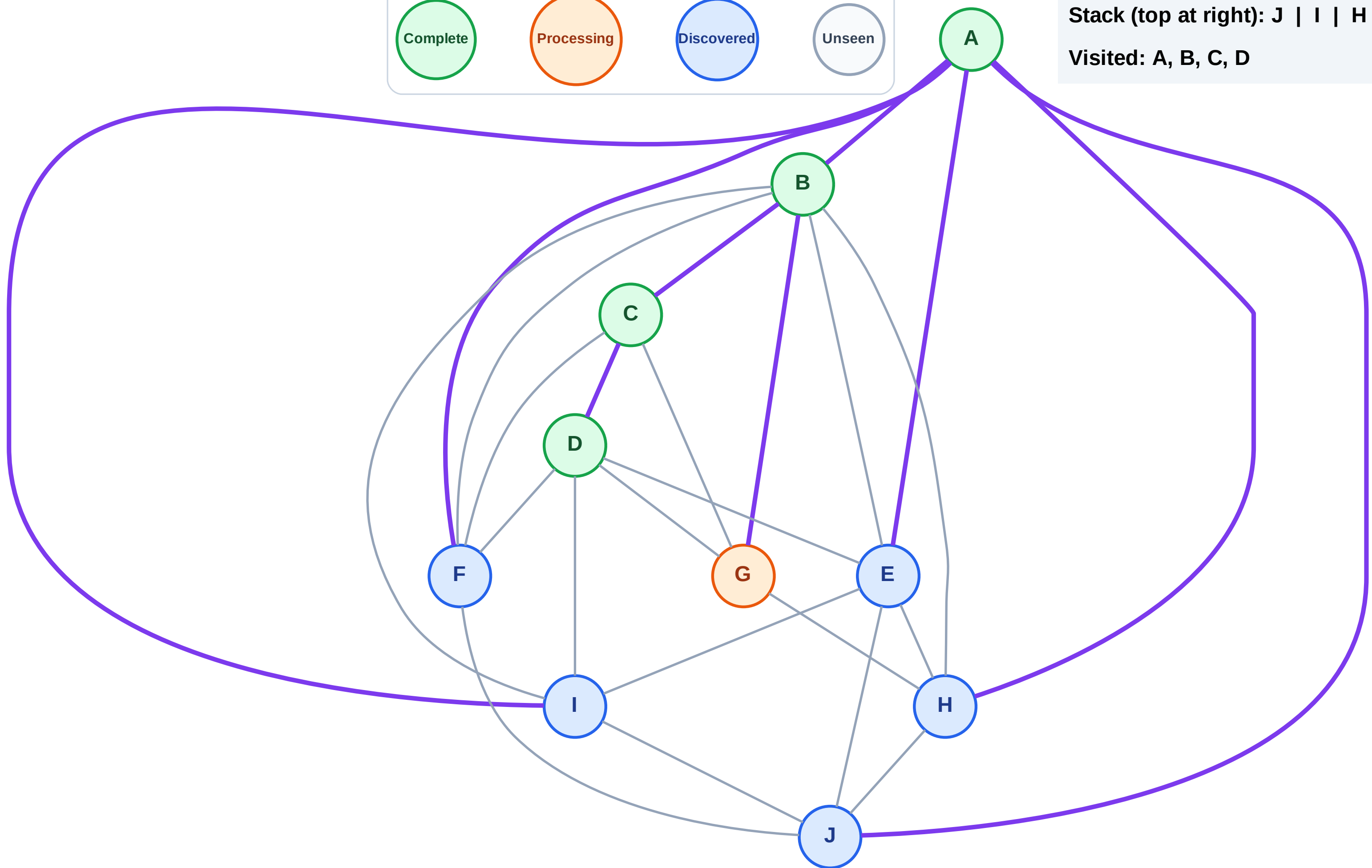
Discovered



Unseen

Stack (top at right): J | I | H | F | E

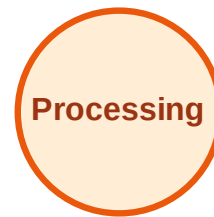
Visited: A, B, C, D



DFS Traversal

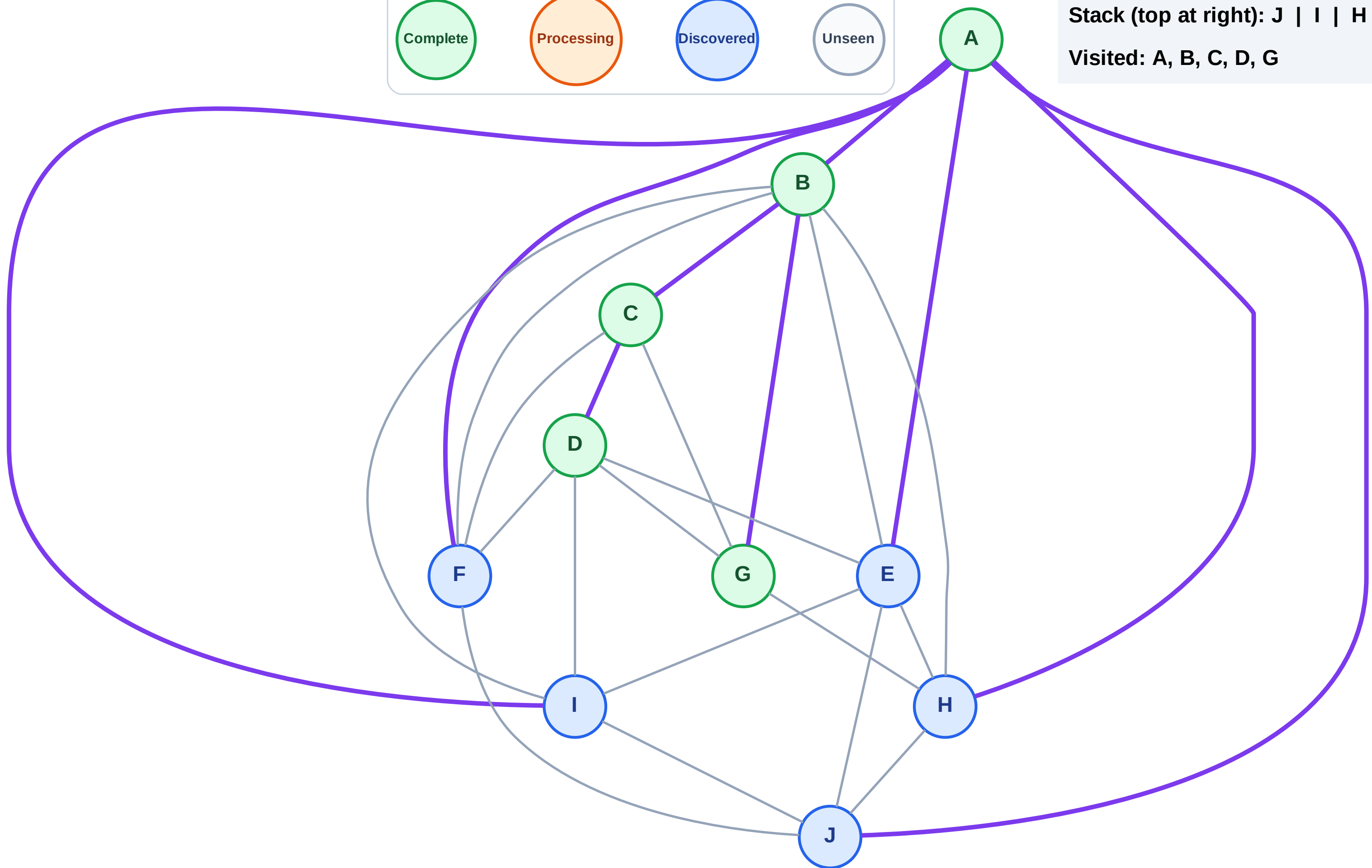
Step 11: No new neighbors from G

Legend



Stack (top at right): J | I | H | F | E

Visited: A, B, C, D, G



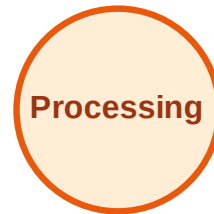
DFS Traversal

Step 12: Process node E

Legend



Complete



Processing



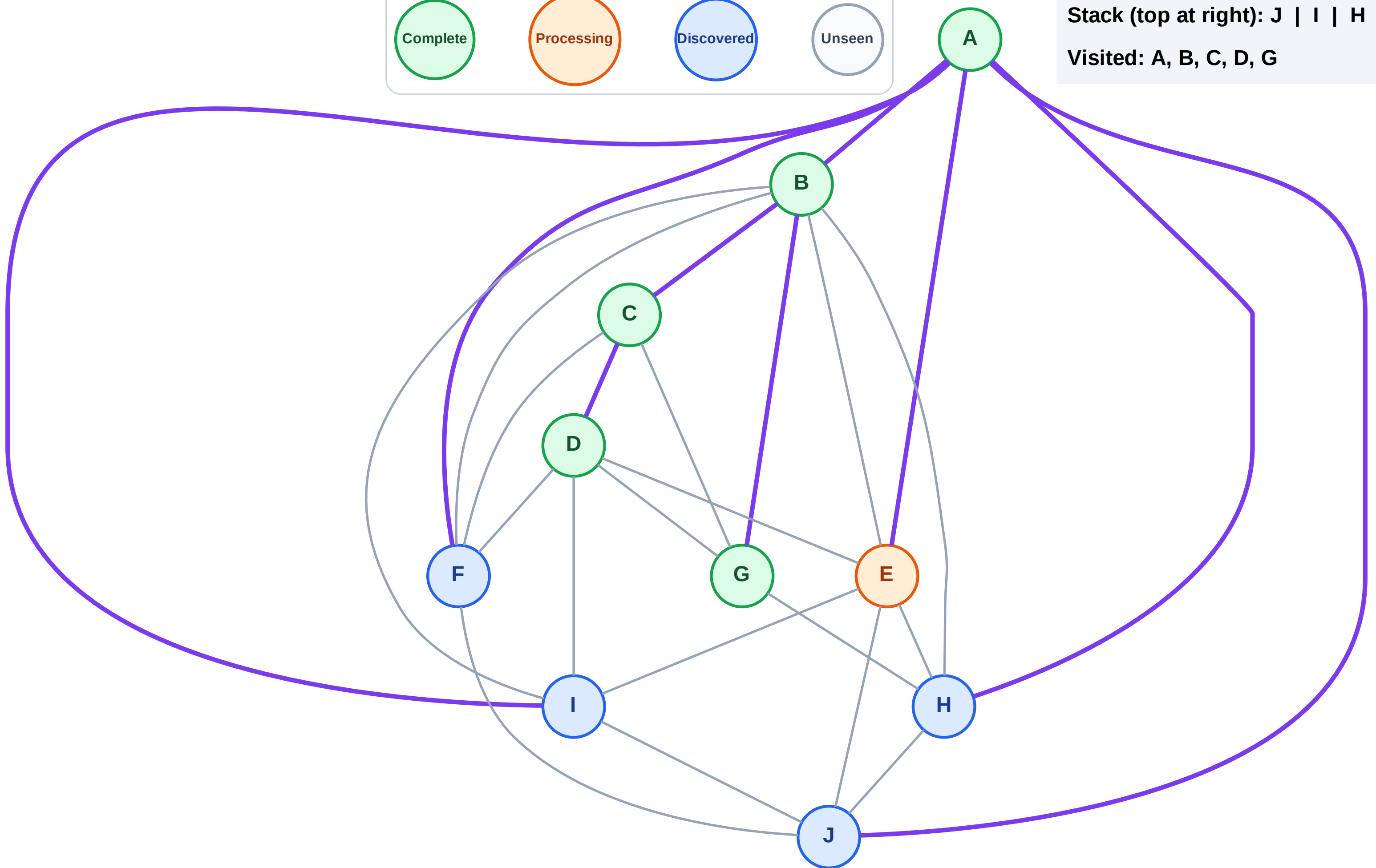
Discovered



Unseen

Stack (top at right): J | I | H | F

Visited: A, B, C, D, G



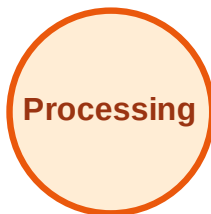
DFS Traversal

Step 13: No new neighbors from E

Legend



Complete



Processing



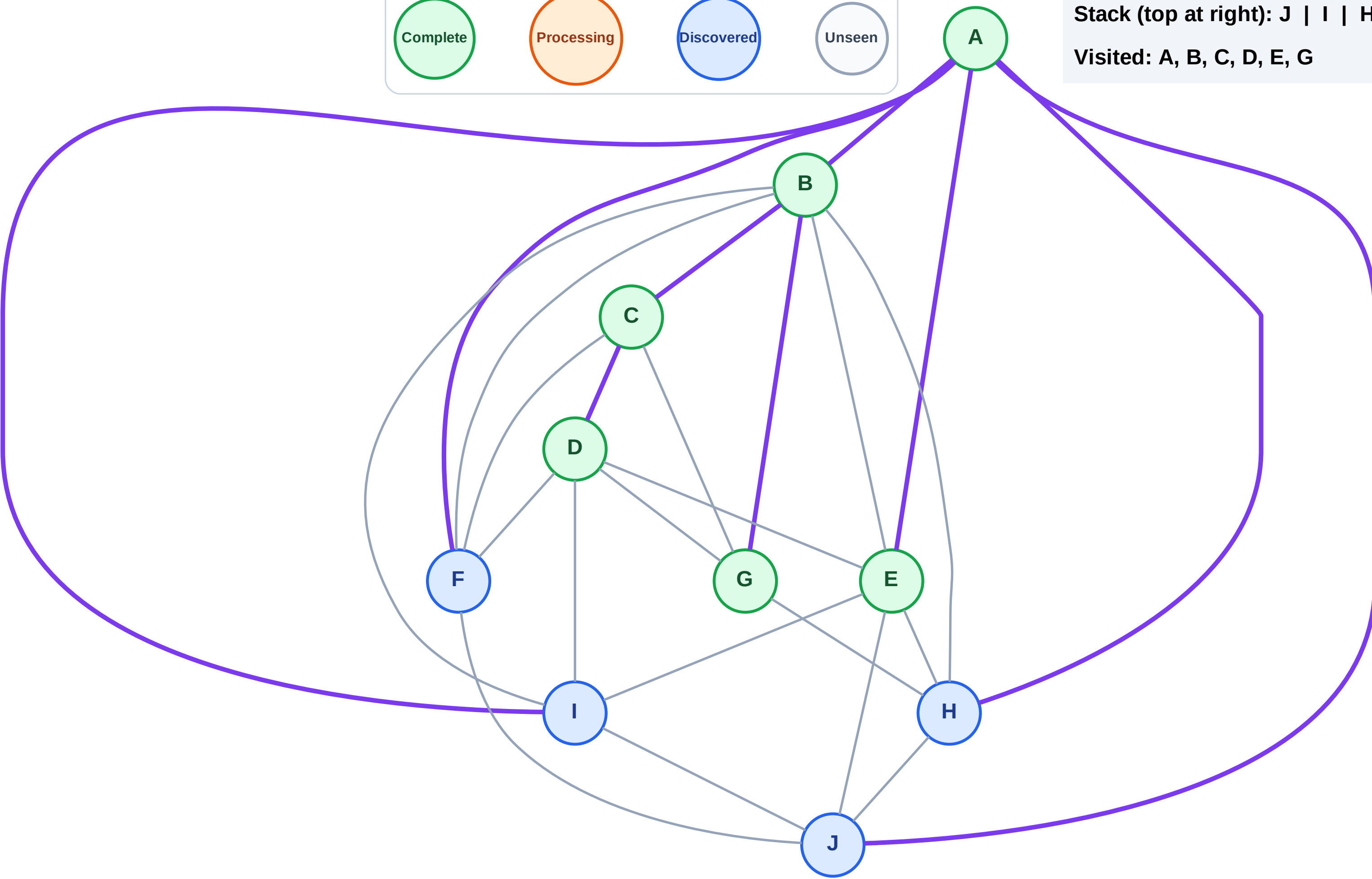
Discovered



Unseen

Stack (top at right): J | I | H | F

Visited: A, B, C, D, E, G



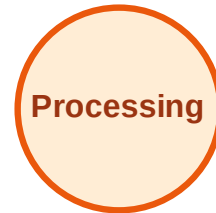
DFS Traversal

Step 14: Process node F

Legend



Complete



Processing



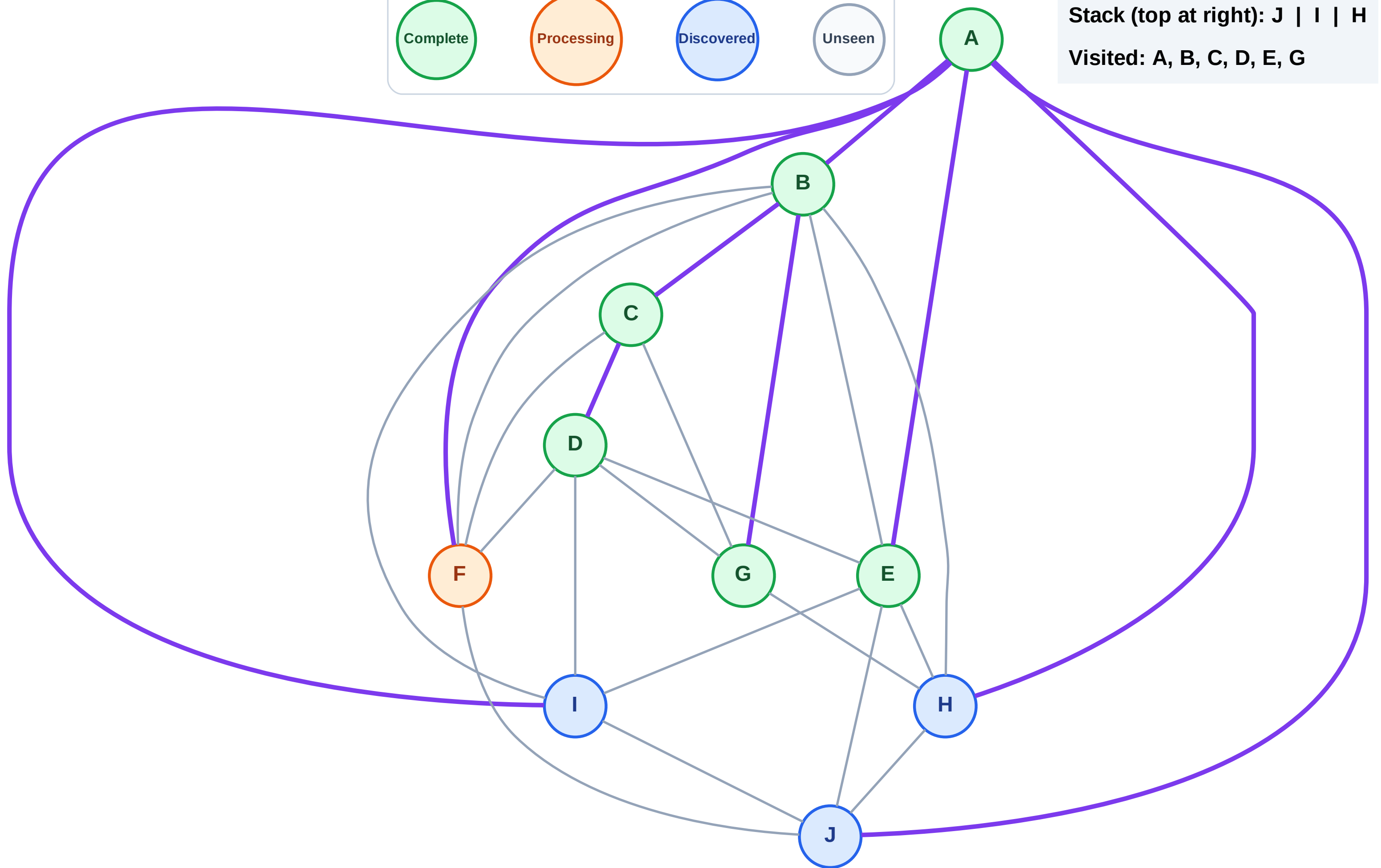
Discovered



Unseen

Stack (top at right): J | I | H

Visited: A, B, C, D, E, G



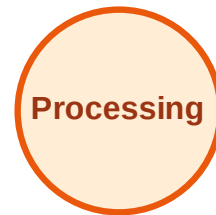
DFS Traversal

Step 15: No new neighbors from F

Legend



Complete



Processing



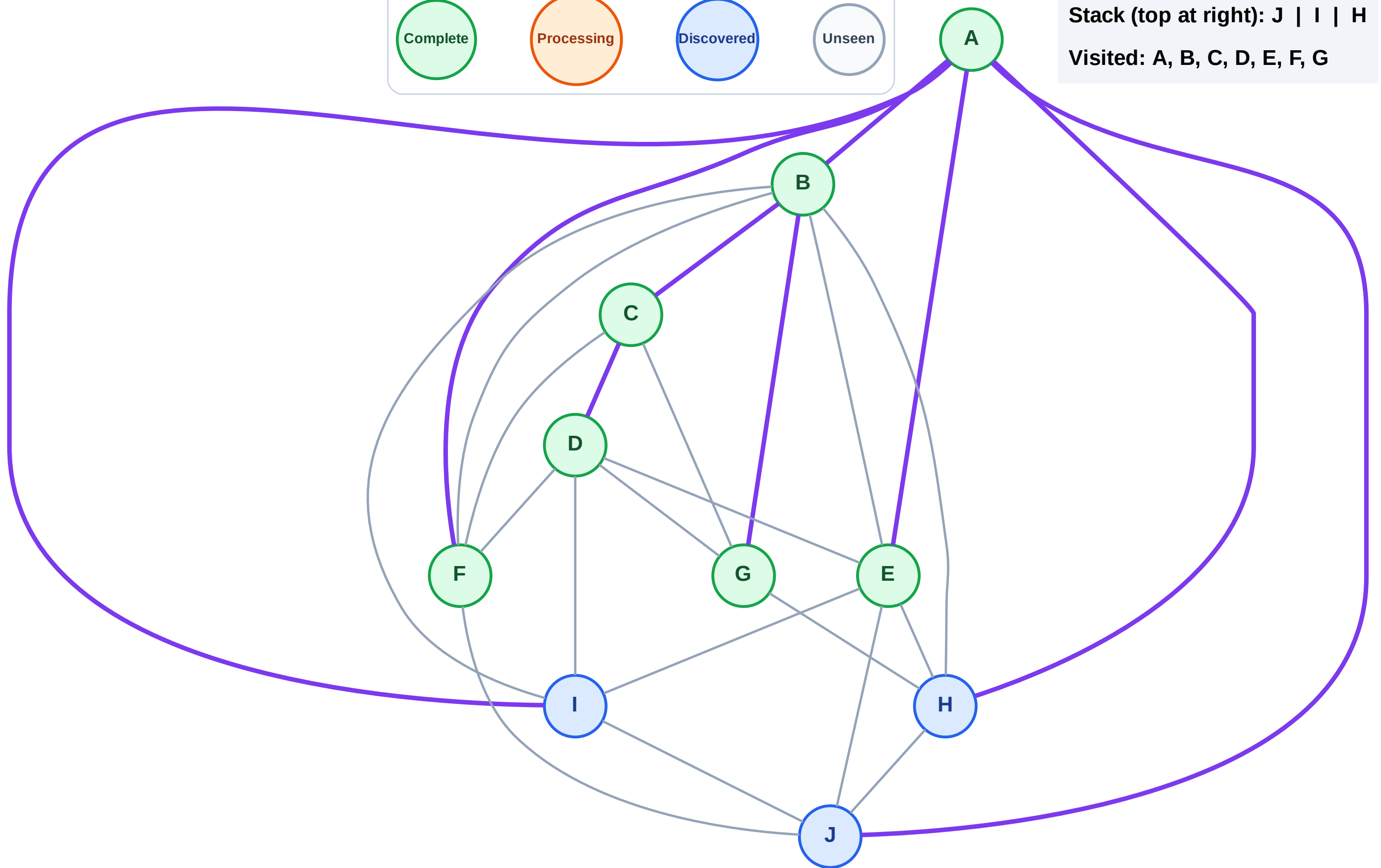
Discovered



Unseen

Stack (top at right): J | I | H

Visited: A, B, C, D, E, F, G



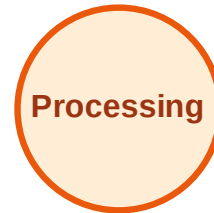
DFS Traversal

Step 16: Process node H

Legend



Complete



Processing



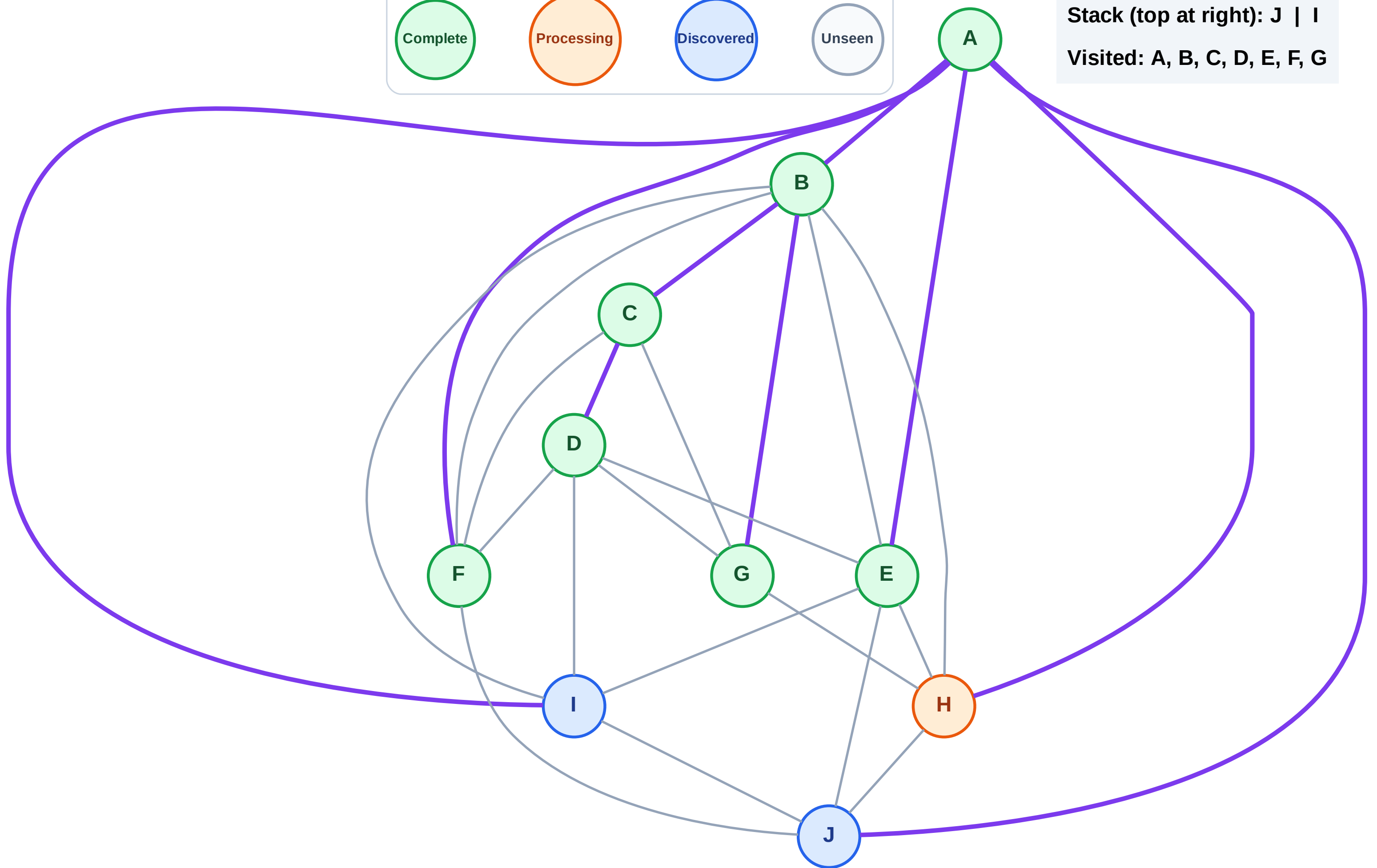
Discovered



Unseen

Stack (top at right): J | I

Visited: A, B, C, D, E, F, G



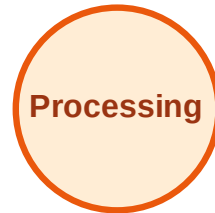
DFS Traversal

Step 17: No new neighbors from H

Legend



Complete



Processing



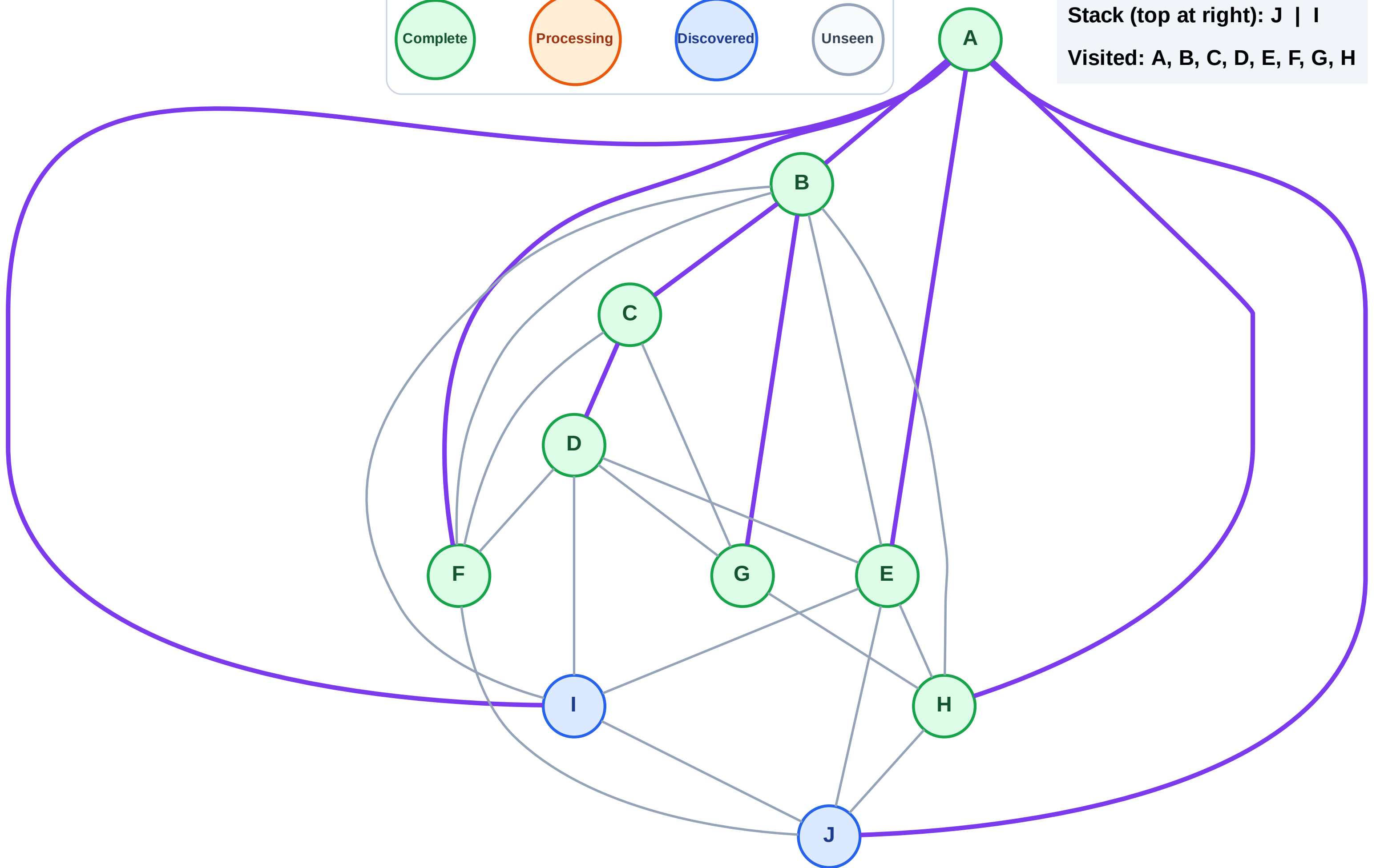
Discovered



Unseen

Stack (top at right): J | I

Visited: A, B, C, D, E, F, G, H



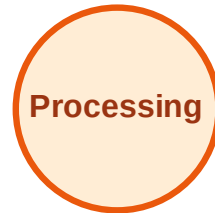
DFS Traversal

Step 18: Process node I

Legend



Complete



Processing



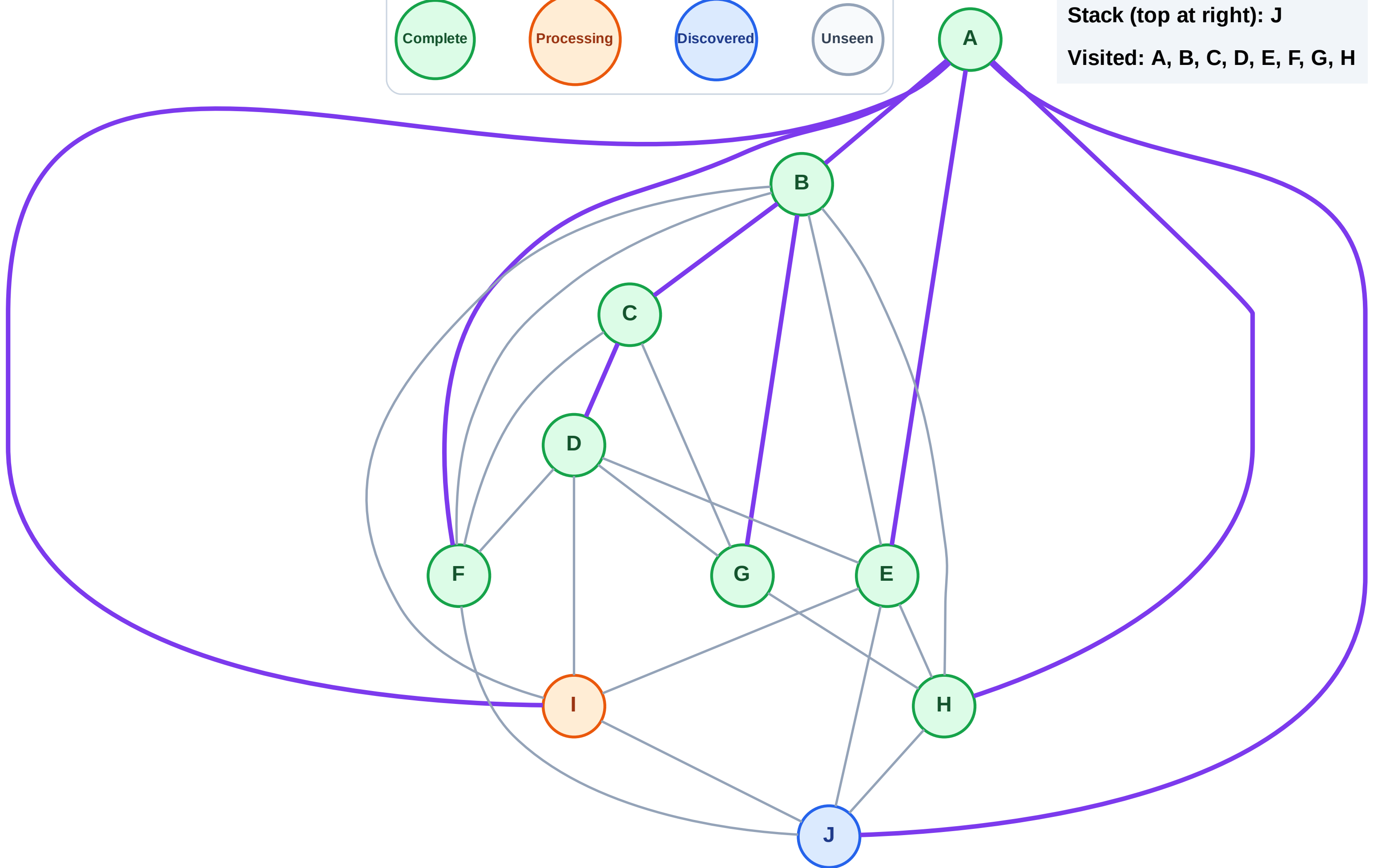
Discovered



Unseen

Stack (top at right): J

Visited: A, B, C, D, E, F, G, H



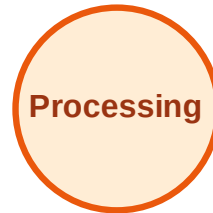
DFS Traversal

Step 19: No new neighbors from I

Legend



Complete



Processing



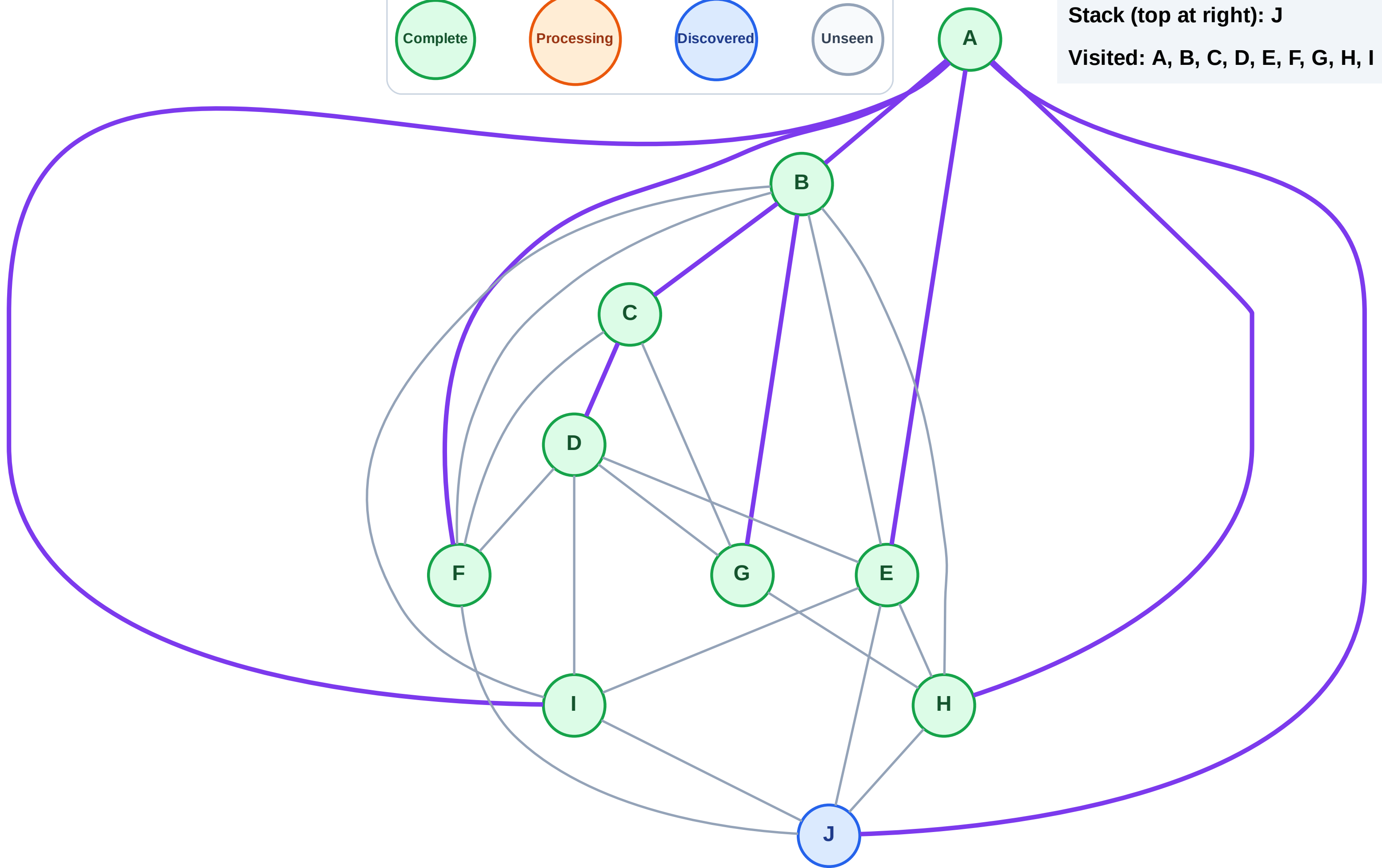
Discovered



Unseen

Stack (top at right): J

Visited: A, B, C, D, E, F, G, H, I



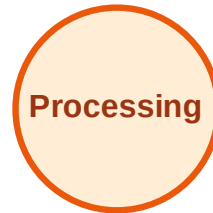
DFS Traversal

Step 20: Process node J

Legend



Complete



Processing

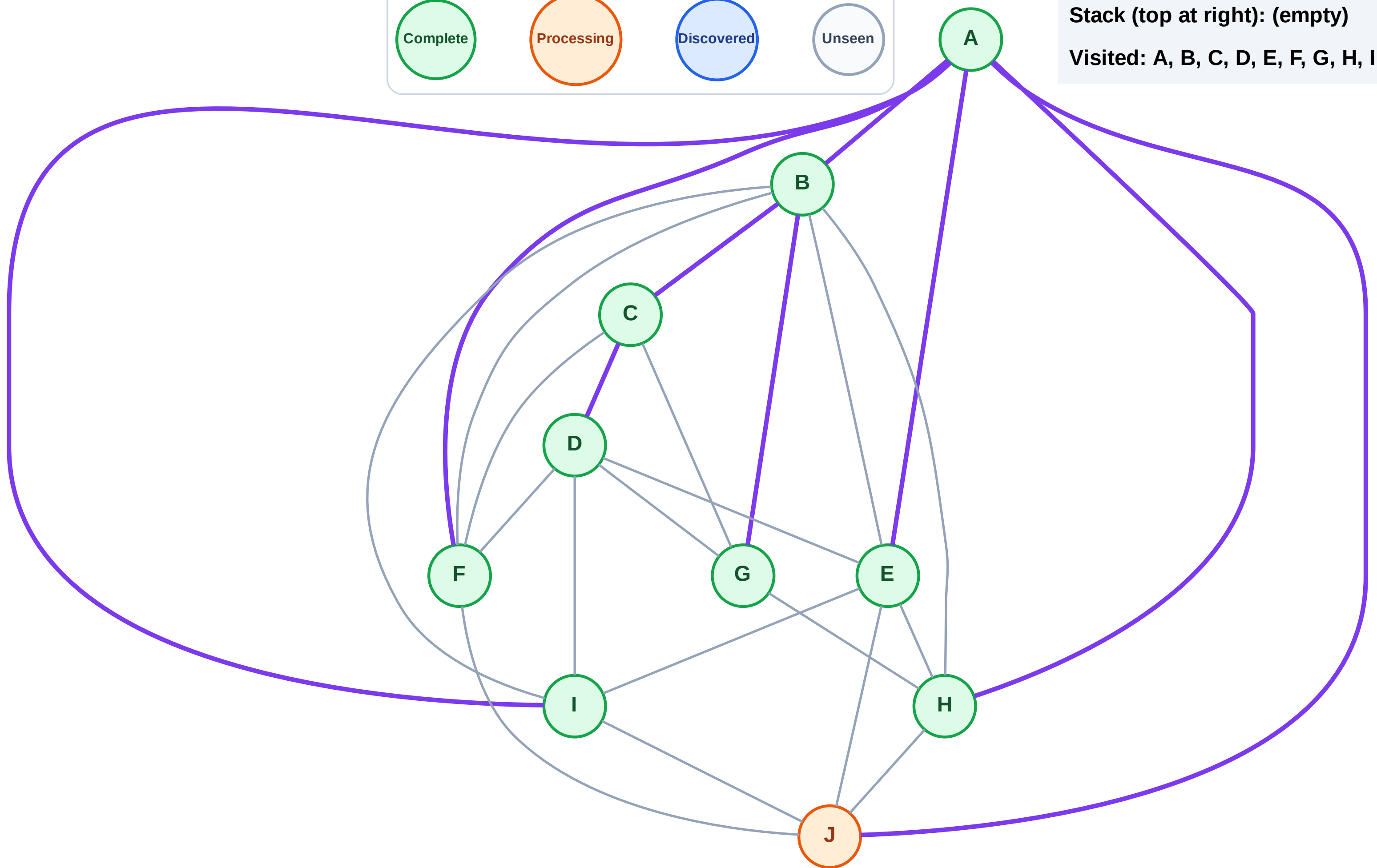


Discovered



Unseen

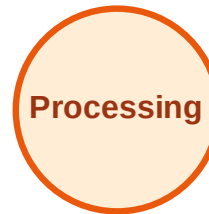
Stack (top at right): (empty)
Visited: A, B, C, D, E, F, G, H, I



DFS Traversal

Step 21: No new neighbors from J

Legend



Stack (top at right): (empty)
Visited: A, B, C, D, E, F, G, H, I, J

